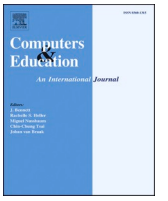
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Dynamics of self-regulated learning: The effectiveness of students' strategies across course periods

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ABSTRACT

Proper self-regulating skills are essential in the new reality of digital learning in higher education. Research has shown that the trace data of students' learning management system activity can identify various online learning tactics and strategies, but also their transitional dynamics, which are linked to academic performance. This study builds on this work by examining how learning tactics and strategies change within individual courses and how this relates to academic performance. A substantial dataset of 41 courses over two academic years at one university is analyzed. Employing Markov models on trace data, we identify prevalent tactics and strategies students use throughout courses. Our study examines shifts in strategy usage, comparing patterns between the initial and latter stages of the courses. The results reveal distinct clusters of learning strategies and their impact on academic achievement. Notably, deep learning strategies show significantly superior performance to surface approaches, especially when maintained over time. Moreover, students who consistently apply the same strategy score higher than those who are inconsistent. However, consistent surface learners score significantly lower than inconsistent learners. Underscoring such longitudinal trends could help interventions, aiding educators in targeting students with weaker strategies at specific times to boost their effectiveness and efficiency. This research contributes to a nuanced understanding of self-regulated learning behaviors in online educational contexts by showing the importance of dynamic transition of learning strategies for educators, instructional designers, and policymakers to enhance student learning experiences and outcomes.

1. Introduction

Students must possess and successfully apply self-regulated learning (SRL) skills to use online learning platforms effectively and achieve learning goals (Winne & Hadwin, 1998). The selection and application of proper learning tactics and strategies are essential self-regulating abilities that can lead to substantial differences in academic performance (Matcha, Gašević, Ahmad Uzir, Gašević, Matcha, Jovanović, & Pardo, 2020; Saint, Whitelock-Wainwright, Gasevic, & Pardo, 2020; Saqr, López-Pernas, Jovanović, & Gašević, 2023). Students engaged in SRL processes review their learning outcomes and relate them to their original learning goals, which leads

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to adaptation in their learning behavior (e.g., learning tactics and strategies) or expectations (Winne & Hadwin, 1998). Understanding the superiority of specific learning tactics and strategies and how students use them is a valuable insight, as this knowledge can be used to improve academic performance through interventions on specific learners' behavior and indirectly through changes in the learning environment, helping students adopt more effective learning behavior.

Traditionally, detecting tactics (e.g., memorizing, highlighting) and strategies (e.g., deep and surface learning) of students relied upon self-reporting measures (Panadero, Klug, & Järvelä, 2016); these come with their own shortcomings, such as the subjectivity of the learner when reporting. Furthermore, Hadwin, Nesbit, Jamieson-Noel, Code, and Winne (2007) found that students are generally inaccurate when reporting their learning behavior. Given that much of education at least partly takes place in online Learning Management Systems (LMSs), it makes sense to consider whether the students' clicking behavior within LMSs can be leveraged to successfully identify learning tactics and strategies. Recent and even some earlier studies suggest this might be feasible (e.g., Fincham, Gasevic, Jovanović, & Pardo, 2019; Jovanović, Gašević, Dawson, Pardo, & Mirriahi, 2017; Saqr et al., 2023a, b, c).

More advanced analytical techniques have allowed researchers to contextualize the data, which is crucial for understanding SRL and getting more in-depth interpretations of students' online learning behavior (Fan et al., 2022). This approach aims to find sequences of clicks that can be interpreted as coherent learning behaviors ('tactics') and to determine sets of coherent tactics that can be interpreted as belonging to learning strategies ('strategies'). Studies have looked at various time partitions, from weeks (e.g., Matcha, Gašević, Uzir, Jovanović, & Pardo, 2019, pp. 461–470; Villalobos, Pérez-Sanagustín, & Broisin, 2024) to entire courses (e.g., Uzir et al., 2020) or even entire programs (e.g., Saqr et al., 2023a, b, c). However, irrespective of the chosen scale, it is important to consider that the learning process itself is gradual: tactics and strategies can change over time, or students may adopt different strategies as courses or sets of courses progress (Cristea, Snijders, Matzat, & Kleingeld, 2023; Panadero et al., 2016; Saint, Fan, Gašević, & Pardo, 2022; Saqr and López-Pernas 2021). Instead of focusing on which learning approach students use and which is performing best, it is useful to consider the effect of the transitions between learning approaches on academic achievement. This kind of research is scarce (Saqr et al., 2023a, b, c), especially when it comes to transitions between surface and deep learning strategies at various course periods, partly because it necessitates a relatively large sample of courses and students to identify transitions properly.

This paper aims to enhance our understanding of the significance of learning strategy transitions by examining students' strategies at the course level and their dynamic transitions within smaller course segments. Specifically, we identify the tactics and strategies employed by students, then divide the courses into halves and quarters to investigate how strategies evolve between these segments. We compare these different levels of granularity by analyzing their tactic composition and their relationship to final grades. Our data encompasses 41 bachelor's courses at a STEM university, covering two academic years, representing a notably large scale in the study of strategy changes. Additionally, we will index all study-related actions within the LMS, providing a comprehensive view of students' learning behaviors and their relevance. This leads to the following research questions:

- RQ1: Which learning tactics and strategies of students can be identified through the analysis of the LMS's trace data?
- RQ2: Which strategy transitions can be distinguished between the halves and quarters of the course, and what characteristics define these transitions?

Finally, our investigation explores the impact of transitioning between learning strategies on academic achievement. Previous studies suggest a correlation between strategy transitions and academic performance (e.g., Fan et al., 2022; Saqr & López-Pernas, 2021a; Saqr et al., 2023a, b, c). Additionally, Hattie and Donoghue (2016) argue that all strategies hold some utility, with weaker strategies potentially serving as stepping stones for the subsequent application of more effective ones, highlighting the significance of strategy order.

- RQ3: What is the relation between the strategies and strategy transitions on the one hand and final course grades on the other hand, and what are the consequences of maintaining the same strategy versus altering it?

We postulate that adopting a dynamic perspective in learning, as addressed by these research questions, can offer valuable insights into students' learning in online environments. This approach not only sheds light on the significance of strategies but also underscores the importance of their sequential application.

The following section summarizes the scientific literature regarding the significance of online tactics and strategies and briefly outlines how these are typically extracted from trace data. The subsequent section emphasizes the importance of temporality in such studies and reviews previous research on strategy transitions. Following this, we describe the methodology employed, present the results, and offer a discussion along with suggestions for future research.

2. Related work

2.1. Detecting online learning tactics and strategies

Assessing learning behaviors is key to a substantial part of educational research (Winne & Hadwin, 1998). The correct application of SRL, for instance, has been directly associated with higher academic performance (Broadbent & Poon, 2015). In earlier and current research, learning behaviors were measured based on think-aloud protocols and other self-reporting methods, such as surveys (Matcha et al., 2019, pp. 461–470). Such data collection methods allow researchers to design questions that directly address their specific research objectives, making the data collected closely aligned with what they aim to measure. However, these methods are also subject

to various biases and interpretation issues that can affect data validity and reliability. Their obtrusive nature might directly influence learning behavior through cognitive overload and lead to low ecological validity, and usually, the time and effort involved in data collection are high. Another limitation that is sometimes mentioned is that due to their frequency-based disposition, the statistical methods employed are limited in their findings and interpretations and lead to "ontological flatness" (Bannert, Reimann, & Sonnenberg, 2014). Instead, trace data and process mining techniques allow for unobtrusive collection and more complex analyses that include temporal and sequential approaches, potentially revealing patterns in behavior that are impossible to notice with other methods (Saint et al., 2022). The primary drawback of trace data is that the students' recorded clicks often fall short of immediately representing interpretable behavior. This underscores the point that, although trace data offers certain advantages, integrating both obtrusive and unobtrusive methodologies is essential and should be tailored to the study's specific objectives.

Studies that use trace data to find students' learning tactics and strategies follow a now well-established methodology. However, the terminology has been used somewhat inconsistently, where the terms 'learning tactics' and 'learning strategies' are sometimes used interchangeably in the literature. We define learning tactics as the cognitive routines or techniques students perform toward a certain goal (Derry, 1989; Saqr et al., 2023a, b, c) and the procedure mentioned above generates such kinds of routines or techniques. Students summarizing a text or reading the forum to better understand the homework are both learning tactics. Tactics can vary and heavily depend on the LMS's functions, course design, or available materials in the course (Matcha, Gašević, Jovanović, et al., 2020). A learning strategy is constructed based on many regularly applied tactics, often selecting and mixing them (Matcha, Gašević, Jovanović, et al., 2020). For example, a learning strategy could be for students to watch video lectures first to gain foundational knowledge, then engage in forum discussions to deepen understanding and clarify doubts, and finally, solve optional exercises to practice and reinforce their learning.

Here, we summarize one of the most common approaches to finding learning tactics and strategies. The consecutive clicks recorded in the trace data are divided into learning sessions based on a chosen temporal gap representing the learner's inactivity period. Within sessions, different kinds of clicks are grouped into learning activities based on their function in the LMS (in our case: Announcements, Discussions, Assignments, Submissions, Learning materials, Mandatory quizzes, Optional quizzes, Metacognition, and Course browsing). Methods such as Markov Modelling (MM) or sequence mining are then used to determine transition probabilities between learning activities. The set of learning activities within sessions can then be further clustered into meaningful learning tactics. Subsequently, the procedure is repeated, but now, by clustering the learning tactics into learning strategies, typically based on agglomerative hierarchical clustering (AHC) or some other clustering method (e.g., K-Means). One of the complications in this line of research is the freedom the researcher has concerning the choice of clustering method and number of clusters, as, in part, this is based on a subjective evaluation of the interpretability and reasonableness of different options. We refer the reader to the Method section for a more detailed explanation of how learning tactics and strategies were created in our case. Generally, researchers use the constructed learning tactics and strategies to, for instance an overview of the different materials used during learning (Li & Tsai, 2017), to research which learning theories students apply (Saint et al., 2021), or to showcase other aspects of learning, such as time management (Uzir et al., 2020).

The idea that the dynamics of learning behaviors plays an important role in effective learning can be traced back to an extensive review study by Hattie and Donoghue (2016). According to their proposed model, effective learning occurs if the strategies are applied at the right moment during the learning cycle. This cycle encompasses three main phases: surface, deep, and transfer learning. Surface learning is usually associated with extrinsic motivation and minimal engagement. In contrast, deep learning sees students interested in the subject and investing time and effort, even when this is not necessarily required of them. In this study, we define surface learning as focusing primarily on meeting course requirements or assessments with minimal engagement beyond what is necessary. This contrasts with deep learning, which involves thorough engagement and understanding. While students in the 'grade-oriented' category may show a strong orientation toward assessments, they do so with a limited depth of engagement, aligning closer with surface learning than strategic learning which is focused on maximizing high grades.

Hattie and Donoghue (2016) argue that both surface and deep learning are important, with the former being necessary for students to successfully apply the latter. Consequently, it is deemed acceptable for students to employ surface learning strategies provided they transition to deep learning strategies in time. Finally, the transfer strategy entails students transferring useful learning strategies between different contexts, such as across courses or time periods. Individual student characteristics and course context can influence the strategies' application. Regardless of the details of this framework, it evidently highlights that proper learning may involve changes in learning tactics and strategies over time (Hattie & Donoghue, 2016; Saqr et al., 2023a, b, c). Unfortunately, these ideas have not yet been tested thoroughly. However, the abovementioned approach using Markov Models and sequence mining on students' clickstream data enables such an analysis of transitions and their relationships with student performance.

2.2. *Previous studies highlighting the importance of strategy transitions*

Researchers understand online SRL as a series of dynamic events that cyclically unfold over time (Butler & Winne, 1995). Temporal analysis is necessary to investigate the order, transition rates, or sequence patterns while looking at various levels of time, such as single actions or larger knowledge construction phases (Chen, Knight, & Wise, 2018). We highlight several studies that have used such an approach.

Jovanović et al. (2017) clustered data from one flipped classroom engineering course into four learning tactics and five strategies. When interpreting the strategy transitions, the researchers used the median number of sessions and how these changed between weeks. Overall, the students changed their behavior over time, which is expected (Winne & Hadwin, 1998). Interestingly, they adopted more ineffective strategies towards the end of the course (such as applying more formative but less summative assessment strategies).

Fincham et al. (2019) used three years of data from the same course that Jovanović et al. (2017) had used, with a crucial difference: they clustered the tactics into strategies for each course half separately. They identified eight tactics, which resulted in four strategies for each half of the course. The learning strategies that students adopted in either the first or second half of the course were associated with final exam performance. This association was present regardless of which half the strategy is primarily applied in, with strategies involving consistent, deep engagement linked to higher performance, and those involving surface engagement linked to lower performance. A feedback intervention during the second and third year led students to adopt deep learning strategies more often, increasing their academic performance overall. Using Massive Open Online Courses (MOOC) data, Shirvani Boroujeni and Dillenbourg (2019) identified different longitudinal profiles in online learning patterns. In general, most students demonstrated stability in their choice of strategy: 53% of students maintained their strategies across the six assessment periods. The other half changed strategies over the six periods, albeit not in a consistent way.

Saqr and López-Pernas (2021) used data from 15 courses to identify the longitudinal trajectories of online student engagement across courses at the program level. Three engagement levels were identified at the course level: Active, Average, and Disengaged. These were further clustered, resulting in three trajectories at the program level: Mostly-engaged, Intermediate, and Troubled. The Mostly-engaged students scored significantly higher grades when compared to the latter two. Overall, students had a high probability of not changing trajectories, suggesting that students were mostly consistent regarding transitions across courses. Finally, the less engaged students had the highest chance of transitioning and were more likely to "ascend" to a better, more engaged level. In another longitudinal study, Saqr and López-Pernas (2022) used data from ten successive courses deployed over four years to investigate students' roles in online collaborative learning and the roles' longitudinal evolution. The research found three roles at the course level (Leader, Mediators, and Isolates), which further clustered into three trajectories across all levels (Intense, Fluctuating, and Wallowing-in-the-mire). Similar to their previous study, they found that these roles were rigid as most "re-emerged" and were predictable throughout the program, regardless of the task or course. However, in this case, the most problematic role (Wallowing-in--the-mire) was the most stable. Saqr et al. (2023a, b, c) then investigated longitudinal student behaviors used in two problem-based learning medical courses. They identified four student strategies at the course level (Light interactive, Moderate interactive, Light diverse, and Intense diverse), resulting in three program-level trajectories (Interactive-intense, Light, and Intense). Overall, students with Light engagement scored significantly lower grades than students with Interactive-intense engagement. However, only a small group adopted Interactive-intense behaviors, and an even smaller subgroup successfully transferred them to a new course. Students who used Interactive-intense engagement obtained higher grades and were less likely to adopt surface learning strategies.

Overall, the literature suggests that there might be a relation between how students transition between the learning strategies applied in various course periods and the final grades. Most often, students tend to keep their strategies the same throughout courses, making their initially adopted strategy very important for academic performance. Indeed, Shirvani Boroujeni and Dillenbourg (2019) found that the consistent students scored higher than students changing strategies. Their study indicates consistency in study patterns, especially those involving regular engagement with course materials, benefits learners. Specifically, patterns that involve consistent video watching and assignment submission after a few inactive days tend to result in better performance. When looking at the engagement of students, Saqr and López-Pernas (2021b) found that transitions between engagement states were more frequent in less engaged students. For instance, Disengaged students were more likely to ascend to Average engagement states (39%) and Average students to Active states (20%). This suggests that while some level of consistency is beneficial, the ability to transition to higher engagement states can lead to improved outcomes. In a full program study (Saqr et al., 2023a, b, c), look ed at the longitudinal relationship between student engagement and achievement. They found that this relation varies significantly over time and across subgroups, however, general consistency in engagement was important for performance. For instance, high-achieving students tended to maintain their performance despite occasional dips in engagement, suggesting stability in their learning strategies. In contrast, students with lower initial engagement often saw persistently lower achievement and were more prone to drop out if they did not increase their engagement. Transitions between engagement levels were particularly important; students who managed to increase their engagement levels over time were more likely to achieve at least moderate success.

In line with this research, Villalobos et al. (2024) examined the evolution of learning strategies over time, using metrics such as the number of active weeks, diversity (capturing the heterogeneity or entropy of weekly strategies), and complexity (capturing the unpredictability or instability of these strategies). Their findings indicate that consistent engagement is a stronger predictor of academic performance than any particular sequence of strategy use. This suggests that a stable approach may better support learning by allowing students to build upon previous actions with less cognitive load. Poquet, Jovanovic, and Pardo (2023) similarly highlight the importance of consistency in strategy application. In their study, the researchers use data from an university engineering course to track weekly entropy and form "trajectories of destabilization". This led to three profiles of change: stable weekly study, steep changes in weekly study, and moderate changes in weekly study. They found that high early entropy—or initial variability in study behaviour—was associated with lower academic performance, suggesting these students may struggle with early course demands. In contrast, students with more stable behaviour patterns achieved the highest performance, supporting the idea that maintaining a stable, yet adaptable, study pattern can be advantageous.

Fan et al. (2022) used MOOC data to identify students' learning strategies in a way that stays a bit closer to the definition of SRL. They examined whether students who failed initially would benefit from a change in strategy in subsequent attempts. The researchers identified five learning strategies, which led to eight strategy changes, with only two transitions resulting in significantly higher performance than the no-change groups. In this case, the effects of the consistency of strategy adoption depended on the specific strategies, as most strategy changes did not result in better scores or even led to decreased performance. As evidenced, students engage in various strategies, primarily categorized as surface or deep learning, leading to observable transitions in strategy employment, an essential component of SRL referred to as study adaptation (Jovanović et al., 2017).

In this study, we aim to explore these incongruences and expand upon previous research on online strategy transition using pattern analysis methods, similar to the methods used in the Fan et al. (2022) and Saqr and colleagues' studies. When extracting tactics and strategies, as much as possible we try to distinguish students based on behaviors with meaningful content, as opposed to mainly based on intensity. The fact that we use a set of courses that is an order of magnitude larger than what is common will more easily allow us to consider how strategy transitions can provide insights into student learning behavior and how they connect to academic achievement. In line with Hattie and Donoghue (2016), we acknowledge that surface learning strategies may not inherently be detrimental, at least not if followed by a transition to deep learning strategies and their subsequent transfer to diverse contexts.

3. Method

3.1. Participants and data

Student clickstream data were collected from the Canvas LMS at a Dutch STEM university across two academic years. The data spanned multiple quartiles, each representing a 10-week period with 8 weeks of instruction and 2 weeks of exams (Table 1). The median number of students per course was Mdn = 138 with a minimum of 17 and maximum of 285. To address our third research question, we gathered the students' final course grades, rated on a scale of 1–10, where a score of 5.5 or higher indicates a student passing the course. The university's ethical review board and the university's Learning Analytics committee approved the study.

The final dataset included 3 056 174 learning events (student clicks) distributed over 401 750 online study sessions from 5870 students in 41 courses. On average, students had M = 68.4 (SD = 40.6) sessions per course lasting on average M = 215.6 s or 3.6 min per session (SD = 446.5 s or 7.4 min); there was an average of M = 9798.8 (SD = 5504.3) sessions per course.

3.2. Data pre-processing

All data pre-processing and analyses were done using R (R Core Team, 2022), using the Saqr et al. (2024) protocol. All the scripts and a short explanation of their usage can be found here: Anonymized Repository - Anonymous GitHub (4open.science). All raw data were first anonymized. Next, clickstream data were cleaned, including removing non-student actions (teachers' clicks or ping requests). The LMS data were connected with the student's final course grades to answer RQ3. The learning patterns were conjugated from the students' online learning activities they generated using Canvas. The next step was to decide on the clicks/events we would use. For this, we identified the most used features and learning activities within the LMS, which led to the creation of 9 large categories (announcement, discussions, assignment, submission, learning materials, mandatory quiz, optional quiz, metacognition, and course browsing). These categories covered more than 99% of the students' clicks in the LMS (besides non-learning-related actions, such as checking the settings, the calendar, or the LMS helpdesk).

The *Announcement* feature is mostly used by educators when they have a general message for the whole course. These are one-way and can carry information regarding assignments, learning material, group formation, or anything else the students need to know about; note that students can also activate the option to receive emails with the announcements, which might cause some students not to access the announcement feature in the LMS. *Discussions* are two-way and are used as forums where students can interact with each other and the educator. The *Assignment* category represents clicks on the assignments page, which holds all the information needed for a student to solve an assignment correctly, such as descriptions and deadlines. After finishing an assignment, the student must submit it, represented by the *Submission* category. This represents the clicks on the actual submission of the final assignment. The distinction is relevant as it can be used for various characteristics, such as determining how early/late the student was or how often the student accessed the assignment information before submitting it.

One of the main functions of current LMSs is being repositories for learning materials. Thus, labeling *Learning materials* is essential. Canvas does not distinguish learning materials from other files within the course, making it difficult to categorize them into subcategories. Ultimately, we used four materials when creating the category: video lectures, lecture slides, other materials (based on file extensions: pdf, doc, csv, etc.), and wiki pages. A teaching mechanism that is often used at universities is the *quiz*. These are used to assess students' knowledge at various points during the course and provide immediate feedback. This is done through Mandatory and Optional quizzes. Since many courses use this and it is a different assessment method than submitting documents, we decided to differentiate them from the submissions label.

Table 1
Course distribution per academic year/quartile and study year.

Quarter	First-Year Courses	Second-Year Courses	Third-Year Courses	Total Courses
2020: Q3	1	3	0	4
2020: Q4	3	3	0	6
2021: Q1	0	2	3	5
2021: Q2	0	2	3	5
2021: Q3	2	3	0	5
2021: Q4	3	3	0	6
2022: Q1	0	2	3	5
2022: Q2	0	2	3	5
Total	9	20	12	41

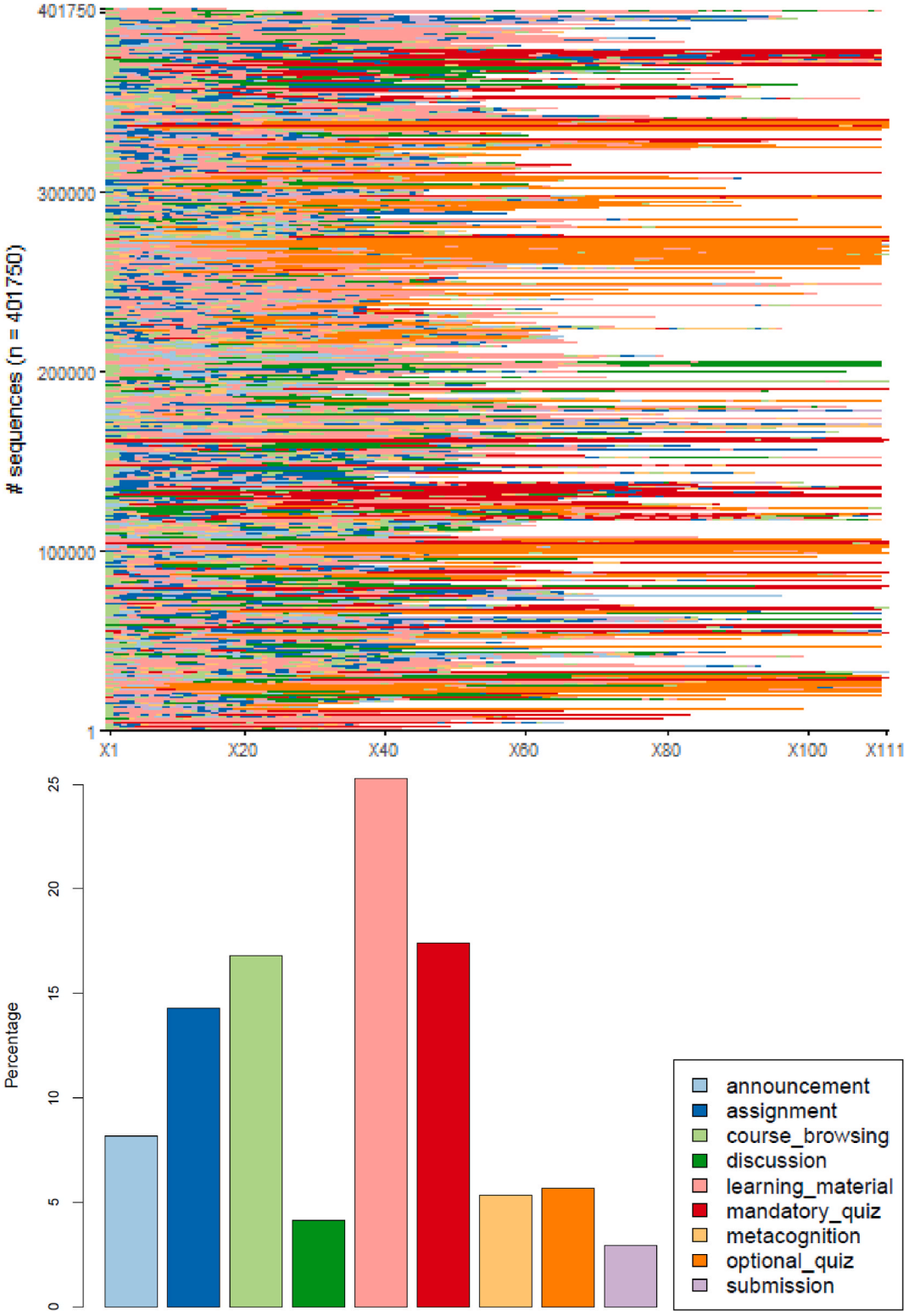


Fig. 1. Index Plot (1a) and Barplot with the Percentages of Actions (1b). The majority of clicks are concentrated on learning materials, which are well-distributed throughout the entirety of the course.

The *Metacognition* label covers all events that represent various support materials, such as the rubrics and assessment matrix, the schedule (the timetable of the course, the course syllabus, the course grade book, and the course student group (if the course has group assignments). Finally, we have the *Course browsing* category, which covers the clicks of students while exploring the various pages of courses, the clicks taking place between all the previous actions, such as clicks when changing the modules. The only clicks not included were non-learning clicks, such as clicks on settings and the general courses-overarching LMS homepage. It is important to mention that not all courses use all the mentioned features. However, no matter the course, these categories cover the vast majority of clicks for each course. Thus, even if applied to a new course, these nine labels would cover most of the online learning activities of a student using Canvas.

To create the learning tactics, we grouped the learning events into sessions. A learning session was defined as a series of activities with at least two clicks within a certain period. As a threshold, we used the 85th percentile of time (725 s or 12.1 min) between consecutive learning events (Saqr & López-Pernas, 2021b). Any gap between two learning events over 12.1 min would start a new session. We continued by calculating the session lengths to understand how frequently students interact with the course material and identify any patterns or anomalies in their behavior. Based on this information, we decided to trim excessively long sessions, i.e., longer than the 90th percentile (Heikkinen, López-Pernas, Malmberg, & Tedre, 2022); thus, a study session can have a maximum of 111 learning events/clicks. This helps to remove outliers and improve the accuracy of subsequent analyses. Finally, we used the TraMineR R package (Gabadinho, Ritschard, Müller, & Studer, 2011) to create a sequence object portraying the learning sessions and acting as input into our analysis. When clustering, we trimmed the tactic sequences to obtain the strategies; this required a different threshold between the different time partitions. The first half had the data comprising the week before the course started and the first four weeks, while the second half had the remaining four study weeks plus the 2 Em weeks and one week after the exams. For the quarter's data, the first quarter comprised the week before the start and the first two weeks, while the other quarters each consisted of three weeks, arranged chronologically. We included only students who had data in all time partitions during the specific analysis (i.e., for the course halves analysis, we only included students who had sessions in both halves). Thus, using the 90th percentile for trimming the sessions, the whole course data had 5870 students with a limit of 120 sessions, the course halves data had 5591 students with a limit of 63 sessions, and the course quarter data had 5110 students with a maximum of 34 sessions.

3.3. Data analysis

To answer the first and second research questions, we ran a simple Markov Model on the sequence object describing the initial probabilities of the learning actions and transition probability matrices between the actions (i.e., the nine Canvas categories described above). These gave us the probability of students starting a session with a specific event (the initial probabilities) and the probability of going from one event to another (the transition probabilities). We followed up by building and fitting Multiple Markov Models (MMM) in a stepwise process fashion (Saqr et al., 2023a, b, c). For the first iteration of the model, we used the clicks of the students as events, with the output being the various tactics (clusters of patterns in learning sessions). We chose the best-fitting model based on the log-likelihood, BIC, and AIC parameters using the expectation maximization (EM) algorithm; each model ran 50 times. We also used various output visualization graphs to ensure we chose the most balanced model with the best interpretation of the student's behavior. The output was used in another MMM model, using the sequence of learning sessions/tactics students employed throughout an entire course to identify student strategy clusters.

For the third research question, we ran non-parametric Kruskal-Wallis tests to check if the various strategies differ in tactic usage and final grades (Kruskal & Wallis, 1952) since the normality assumptions were violated. If significant, we ran Dunn tests (Dunn, 1964) with Bonferroni corrections (Haynes, 2013) to see the pairwise differences. For the course halves data, we formed the Consistent (students who used the same strategy for both) and Inconsistent groups (students who used different strategies); we compared them using a Welch two-sample *t*-test (Welch, 1938). For the course quarters data, we formed the Consistent, Inconsistent, and Undecided groups (students that used all the strategies, a different one per course quarter, without repeating once); here, we used the Kruskal-Wallis followed by the Dunn tests. Finally, we compared the Consistent students within themselves, based on their specific strategy using the same protocol.

4. Results

We first present some descriptive results on the kind and content of the sessions. The longest sessions in terms of clicks usually occur when students fill out quizzes, as seen in the spikes in Fig. 1a; the y-axis portrays the individual sessions, while the x-axis shows the order of clicks within the sessions. As seen in Fig. 1b, Mandatory (17.4%) and Optional quizzes (5.7%) represent a large part of the learning actions with the larger sessions. However, students seem to start a session with another learning action before moving to the quizzes; only 2.7% and 0.9% of initial clicks are on Mandatory and Optional quizzes, respectively. Most of the other actions occur in shorter sessions. Accessing learning materials represents the most online learning actions students take overall (25.3%); this is also the

second most common way to start a learning session, with 19.8% of first clicks being on Learning materials. Course browsing represents 16.8% of total actions and, as expected, more than half of the initial clicks (53%). Clicks on Assignments (14.3%) or Submissions (2.9%) are present constantly throughout the courses, which is expected considering that students often access the LMS to clarify details of the assignment instructions before working offline on assignments and go back online only to submit their work; these actions are also not a typical first click, with 10.2% and 2%, respectively. Social-related actions such as Discussion forum (4.1%) and Announcements access (8.2%), but also the Metacognition actions (5.3%) represent a smaller chunk of clicks; these are also not a common way to start a learning session (2.8%, 6.8%, and 1.8% respectively).

While the Learning Material action is the most common, the quiz actions are the most obvious in the index plot, most probably due to quizzes requiring longer sessions taking place next to each other (e.g., during exams)

4.1. RQ1: which learning tactics and strategies of students can be identified through the analysis of the LMS's trace data?

4.1.1. Tactics

Based on the log-likelihood, AIC, and BIC parameters and the visual interpretation of the tactics' distribution and action composition, the analysis showed that the best-fitting model had four tactics. Table 2 presents an overview of the eventually defined tactics, and Fig. 2 visualizes the actions per tactic. Appendix A provides further information on the initial and transition probabilities of the actions per tactic. Below, we detail each of the four tactics.

1. **Assignment-focused tactic**: Assignment-focused is the most common tactic and has the second-largest average number of clicks per session and third-longest sessions in terms of time. The dominant action is the click on the Assignment page (28.7%), followed by Mandatory quiz clicks (23.7%). This suggests that most of these sessions are dedicated to various forms of homework, as they address features usually used in assessment and grading. When looking at initial clicks (how these sessions start), we see that students start with Course browsing (30.7%) or Mandatory quizzes (18.9%). The largest transition between different actions (Appendix A) is Assignment→Mandatory quiz (21.2%), followed by Learning material→Mandatory quiz (15%).
2. **Material-focused tactic**: The Material-focused tactic is the second most common, representing 32.9% of all learning sessions; these sessions are the third and second longest in terms of average number of clicks and time, respectively. As expected, Learning material clicks represent more than half of all clicks during these sessions (59.2%), followed by Course browsing (14.9%). Most Material-focused sessions start with clicks on Course browsing (58.4%), Learning material (13.2%), or Assignment (12.9%). These sessions are focused on accessing and using the course material, such as articles, lecture videos, or slides. Since many sessions start by accessing the Assignments page, students may begin by determining what they must study before moving on to the actual learning material in a goal-oriented learning behavior. The highest transition probability is Submission→Assignment (56.9%).
3. **Mixed tactic**: The Mixed tactic is the third most common while being the shortest in terms of both the average number of clicks and time. Compared to the other tactics, this cluster is the most diverse in its composition, being formed out of Course browsing (27.5%), Announcements (21.7%), Metacognition (15%), and Learning material (14.6%), with no clear dominant action; Course browsing is also the most common initial click with 58.6%. The highest transition probability between different actions is Course browsing→Assignment (28.7%). This tactic could be used to get general information about the course (based on the high number of Announcements, Metacognition, and Course browsing clicks) before going into more specific learning behavior.
4. **Quiz-focused tactic**: Quiz-focused tactics are the least common sessions (with less than 5% in total). However, these are by far the longest sessions on average, both in terms of clicks and time. As expected, this cluster is dominated by quizzes, mainly Mandatory quizzes representing 63.9% of all clicks, followed by Optional quizzes (14.1%). Learning material is the third most common click in this cluster; however, the drop in frequency is significant, with only 6% of clicks. The most common way to start a session is with Course browsing (47.1%), followed by Learning material (39.3%). The highest transition is Course browsing→Learning material (34%). These sessions seem to be the most direct and explicit in terms of behavior, with students accessing the LMS to solve quizzes, either as part of mandatory work or optional exercise.

4.1.2. Strategies

Next, we clustered the tactics to obtain the strategies. While the tactics describe students' learning sessions, the strategies describe the kind and order of tactics a student used throughout an entire course (or per course half or course quarter). The best-fitting model for the entire course as a unit of analysis had four strategies. In this section, we describe these strategies, focusing in the subsequent section on the results for course halves and quarters, including how students move between strategies. Table 3 shows the descriptive statistics

Table 2
Descriptive data for the four final tactics.

	Assignment-focused	Material-focused	Mixed	Quiz-focused
Frequency	138 565 (34.5%)	132 257 (32.9%)	113 372 (28.2%)	17 556 (4.4%)
Average clicks per session (SD)	7.8 (12.8)	6.5 (9.4)	6.3 (5.6)	23.4 (27.7)
Average session time in minutes (SD)	3.3 (7.5)	3.5 (6.5)	2.9 (5.2)	11.1 (16.2)
Dominant action	Assignment (28.7%)	Learning material (59.2%)	Course browsing (27.5)	Mandatory quiz (63.9%)
Second dominant action	Mandatory quiz (23.7%)	Course browsing (14.9%)	Announcement (21.7%)	Optional quiz (14.1%)
Most common initial action	Course browsing (30.7%)	Course browsing (58.4%)	Course browsing (58.6%)	Course browsing (47.1%)

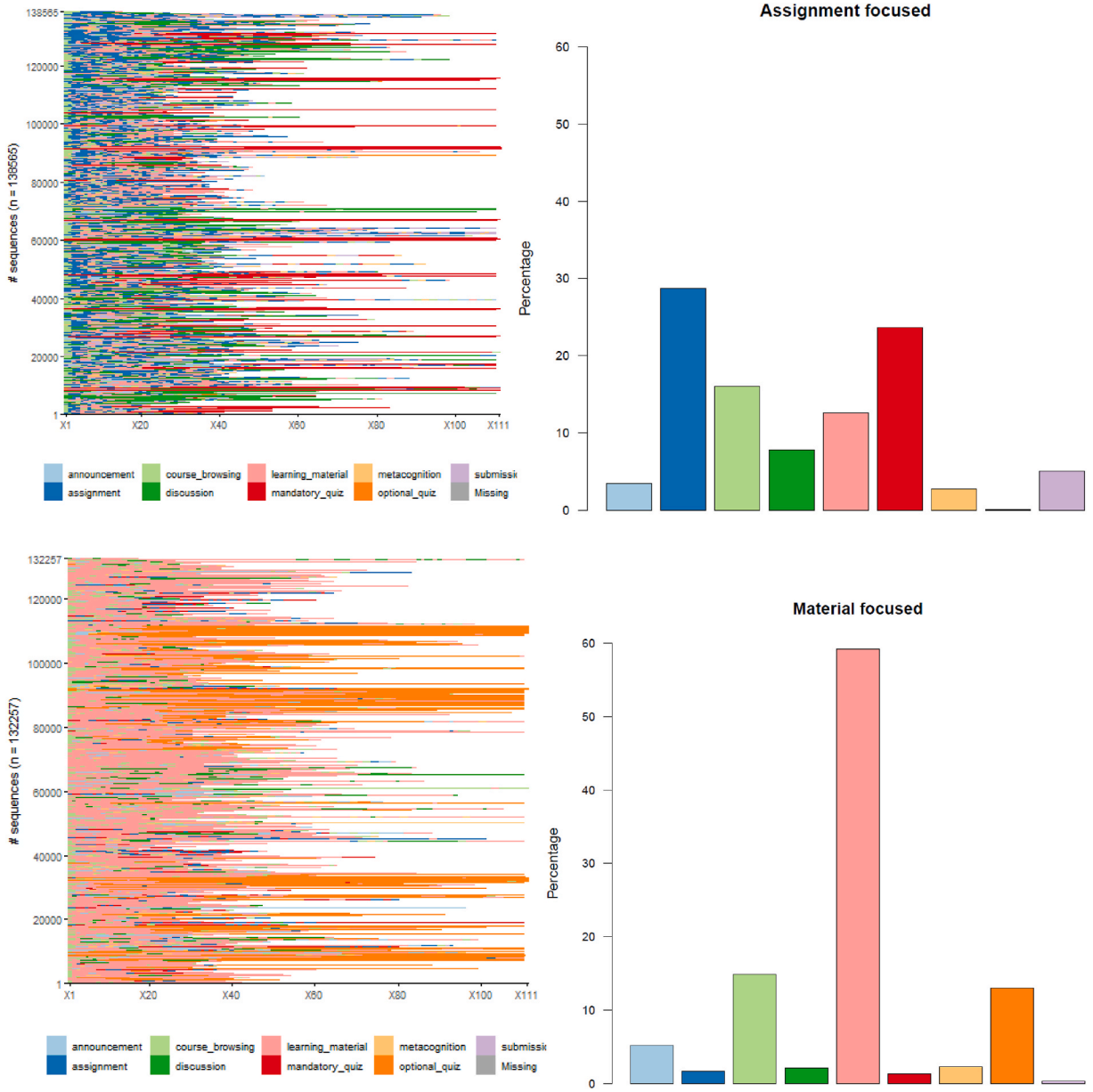


Fig. 2. Index Plots of the Actions (left) and Frequency Plots (right) per Tactic. Mandatory and Optional quizzes require the longest sessions, which tend to cluster together. The other actions mix up and lead to shorter sessions.

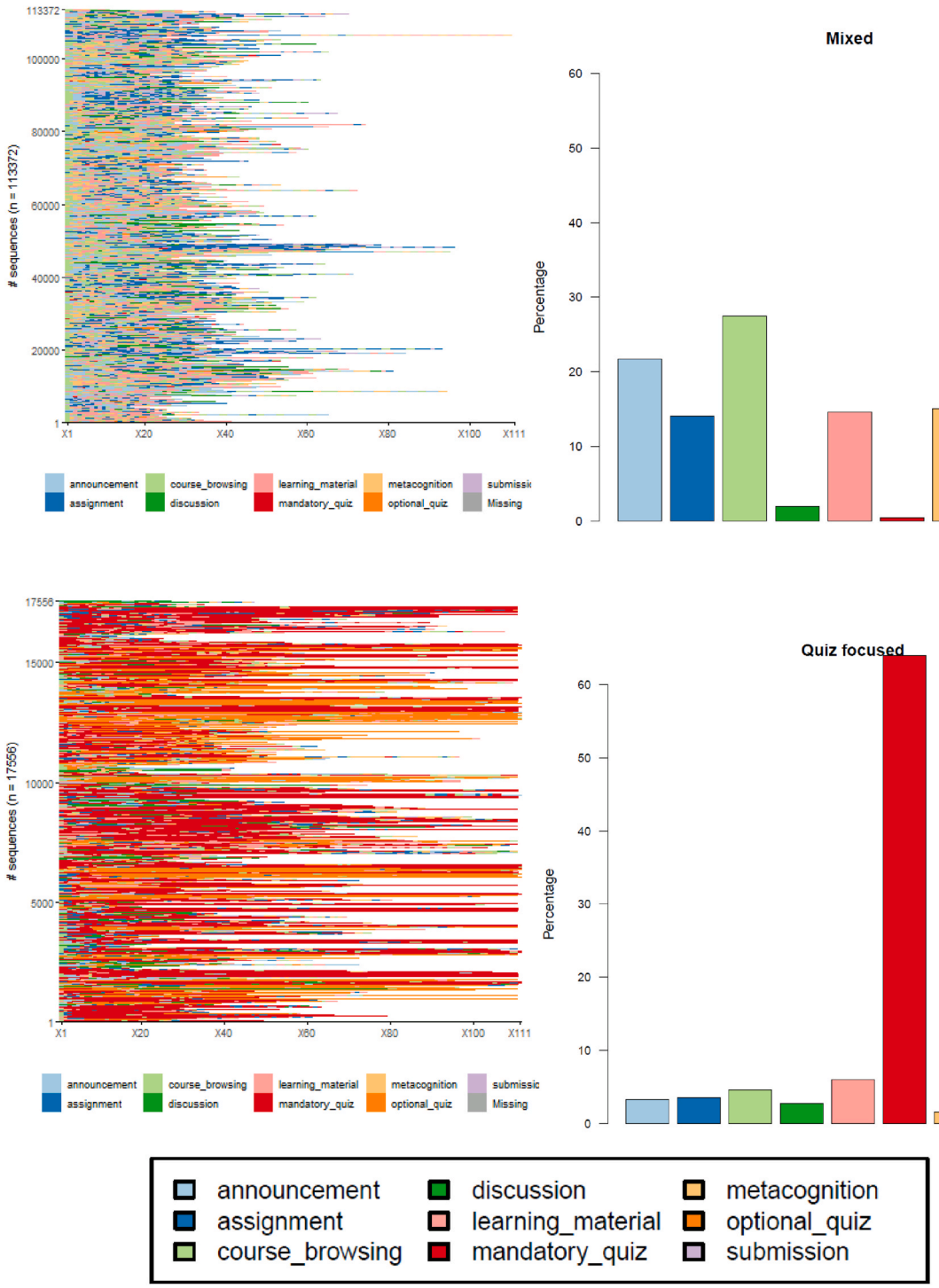


Fig. 2. (*continued*).

for the four strategies found, and Appendix B further specifies the temporal distribution, frequencies, and transition probabilities of the tactics per Strategy. The strategies for the whole course data are:

1. **Balanced strategy**: The first cluster is named Balanced, as it has no clear dominant tactic with three tactics used about equally often. This cluster is the smallest (22.4%) and has the largest average number of sessions per student. The most used tactic is the Assignment-focused (33.3%), closely followed by the Mixed (28.0%), and Material-focused (26.9%). While the Quiz-focused tactic

Table 3
Descriptive data of the four strategies for the whole course.

	Balanced	Resource-oriented	Grade-oriented	Flexible
Frequency	1313 (22.4%)	1461 (24.9%)	1692 (28.8%)	1404 (23.9%)
Average number of sessions per student per course (SD)	76.8 (32.9)	61.3 (31.6)	67 (32.9)	57.2 (32.3)
Assignment-focused	33.3%	18.6%	48.9%	32.8%
Material-focused	26.9	58.4%	26.5%	21.2%
Mixed	28.0%	20.9%	23.4%	44.5%
Quiz-focused	11.8%	2.1%	1.2%	1.5%
Most common first tactic in a course	Mixed (64.3%)	Mixed (59.5%)	Mixed (63.2%)	Mixed (70.5%)

Note. The highlight shows the dominant tactic used in the strategy cluster. The Balanced cluster does not have one. The percentages should be read column-wise as they represent the frequency of tactics per strategy.

represents only around 11.8% of sessions, it is by far the highest percentage compared to the other strategies. The most common way to start a course is with the Mixed tactic (64.3%). Most transition probabilities are around 20%, with the highest being the self-absorption Assignment-focused→Assignment-focused (43%).

2. **Resource-oriented strategy**: Students in this cluster spend most of their sessions accessing the learning resources. The Resource-oriented cluster is the second in size (24.9%) with the third largest average number of sessions per student. This cluster has the highest difference between the tactics used, with 58.4% Material-focused followed by 20.9% Mixed. The most common initial tactic is the Mixed (59.5%), with the self-absorption Material-focused→ Material-focused (69%) being the highest transition probability, followed by Quiz-focused→ Material-focused (47%), and Mixed→Material-focused (45%).
3. **Grade-oriented strategy**: Students in this cluster spend most of their sessions accessing the grade-related materials. This cluster has the largest number of students (28.8%), with the second largest average number of sessions per student. Similar to the Resource-oriented cluster, there is a high difference between the most used tactic (Assignment-focused with 48.9%) and the second (Material-focused with 26.5%). The most common first tactic is Mixed (63.2%), with some considerably high transition

Table 4
Descriptive Data of the Four Strategies for the Course Halves and the Transition Between Them. The "1 → 2" notation represents the change from the first to the second course half or from one quarter to the next.

	Balanced (n= 2527, 22.6%)			Resource-oriented (n= 2975, 26.6%)			Grade-oriented (n= 3251, 29.1%)			Flexible (n= 2429, 21.7%)		
Data	1	2	1→2	1	2	1→2	1	2	1→2	1	2	1→2
Frequency	1208 (21.6 %)	1319 (23.6 %)	+111 (+2%)	1299 (23.2 %)	1676 (30.0 %)	+377 (+6.8 %)	1534 (27.5 %)	1717 (30.7 %)	+183 (+3.2 %)	1550 (27.7 %)	879 (15.7 %)	-671 (-12%)
Average number of sessions per student per course (SD)	39.2 (17.6)	36.3 (16.8)	-2.9 -(1.2)	32.3 (17.7)	35.7 (17.4)	+3.4 -(0.3)	35.6 (16.5)	32.1 (17.5)	-3.5 +(1.0)	31.9 (17.2)	26.9 (16.1)	-5.0 -(1.1)
Assignment-focused	31.9%	33.1%	+1.2%	18.7%	19.2%	+0.5%	51.5%	52.0%	+0.5%	31.7%	31.0%	-0.7%
Material-focused	25.1%	25.3%	+0.2%	55.8%	60.6%	+4.8%	23.6%	25.4%	+1.8%	20.5%	20.9%	+0.4%
Mixed	30.9%	28.9%	-2.0%	23.3%	17.6%	-5.7%	24.2%	21.0%	-3.2%	47.3%	46.4%	-0.9%
Quiz-focused	12.1%	12.7%	+0.6%	2.2%	2.6%	+0.4%	0.8%	1.6%	+0.8%	0.5%	1.7%	+1.2%
Most common first tactic in a course	Mixed (63.1 %)	Assignment-focused (40.3 %)		Mixed (59.4 %)	Material-focused (39.7 %)		Mixed (61.3 %)	Assignment-focused (70.3 %)		Mixed (73.3 %)	Mixed (49.5 %)	

Note: Green represents an increase, while red represents a decrease between the course halves. The transition between the halves does not have an initial tactic.

probabilities, such as self-absorption of Assignment-focused→Assignment-focused (57%), followed by Quiz-focused→Assignment-focused (46%), and Mixed→ Assignment-focused (45%).

4. **Flexible strategy:** Students in this cluster use various LMS features, allowing them to adapt to courses more easily but not in a balanced way. The Flexible cluster is the third in size (23.9%) while having the smallest average number of sessions per student. Mixed (44.5%) is the most used tactic, followed by Assignment-focused (32.8%). This cluster has the highest percentage of Mixed tactics as initial sessions (70.5%), with the Mixed→Mixed (49%) and Assignment-focused→Assignment-focused (43%) self-absorptions being the highest transition probabilities.

Some of the identified clusters of learning strategies can be conceptually linked to the surface and deep learning strategies paradigm. For instance, the Grade-oriented cluster, primarily centered on accessing grade-related materials, suggests a more surface-level approach to learning, prioritizing extrinsic motivators. Similarly, the Balanced cluster makes high usage of both assignments and quizzes, both used for grading purposes, also suggesting a more surface-oriented approach. On the other hand, the Resource-oriented strategy describes students accessing learning resources and utilizing various features of the LMS to better understand the course material, behavior usually connected to deep learning. The Flexible cluster is more challenging to label, which also shows that it is essential to recognize the subjectivity inherent in labeling these strategies.

4.2. RQ2: which strategy transitions can be distinguished between the halves and quarters of the course, and what characteristics define these transitions?

With the strategies identified, we now focus on the changes in student strategies between the course halves and course quarters. First, we ran the same clustering approach as with the full courses, but now on the course halves and quarters. The first thing to note is that in both these cases, we find the same four clusters as in the full course case, which serves as some evidence for our approach's reliability and makes the interpretation of transitions between clusters less problematic. Based on Table 4, Fig. 3, and Appendix C for the course halves, and Fig. 4 and Appendices D and E for the course quarters, we find the following:

1. **Balanced strategy**:

Halves. The Balanced strategy is the third in frequency (22.6%), with the smallest increase in frequency between halves (+2%). On average, the number of sessions drops by 2.9 from the first to the second half of a course. The most used tactic is Assignment-focused in both halves. The most common way to start a course is with Mixed (63.1%) and Assignment-focused sessions (40.3%) for the first and second half, respectively. This cluster is the most invariable regarding transitions, with all difference probabilities below 5% (Appendix C).

Quarters. Overall, the Balanced cluster is the smallest (19.4%), with an increase of +9.6% between the first two quarters, followed by small fluctuations, making it the most stable when it comes to size. The use of tactics fluctuates throughout the quarters, the biggest

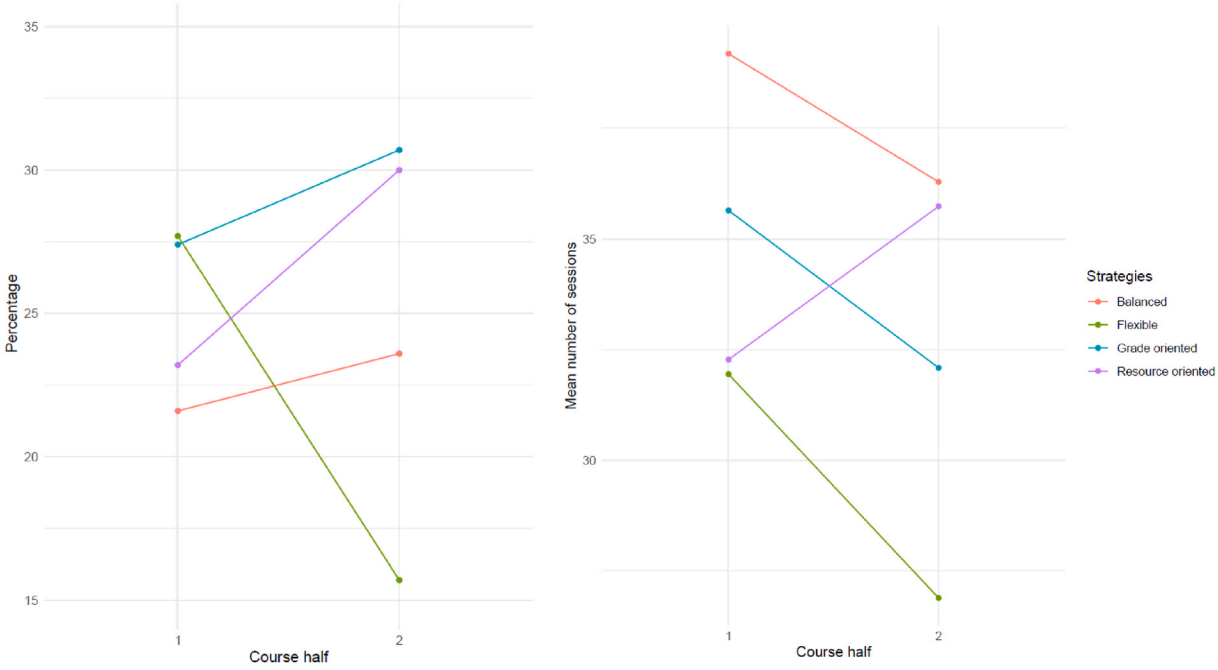


Fig. 3. Line Plot of the percentage (left) and the Average Number of Sessions (right) per Strategy Cluster Between the Course Halves.

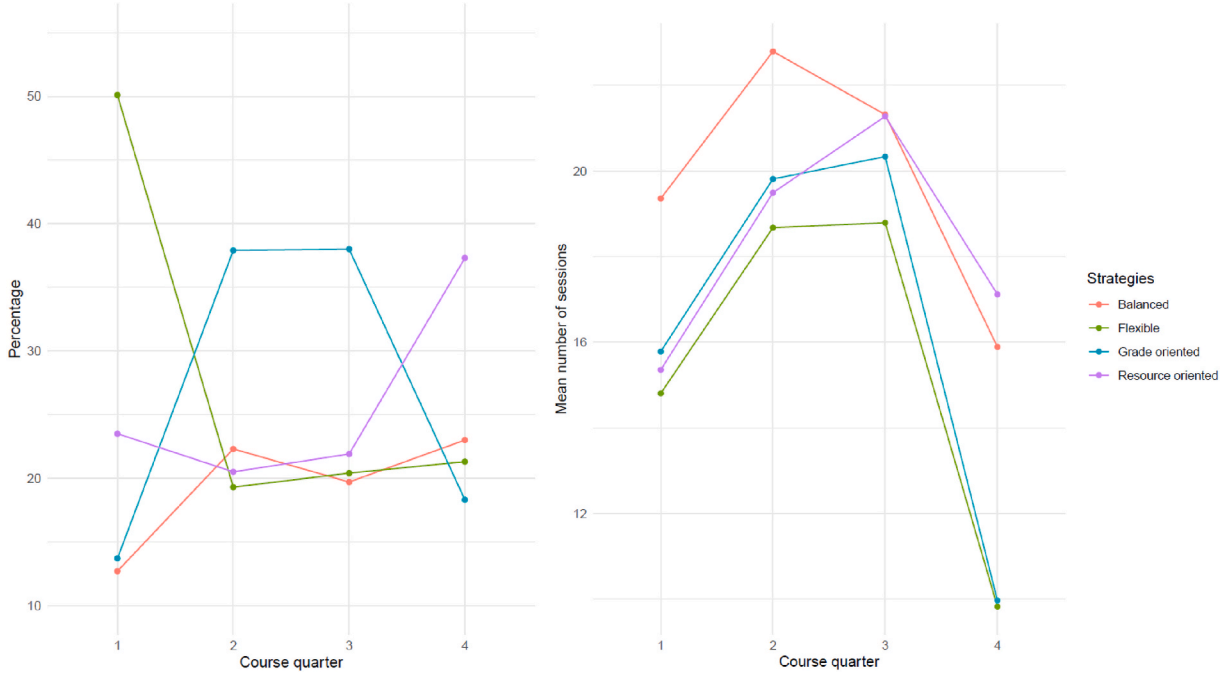


Fig. 4. Line Plot of the percentage (left) and the Average Number of Sessions (right) per Strategy Cluster Between the Course Quarters.

differences occurring for 3 → 4 (−6.7% for Assignment-focused) and during 1 → 2 for Mixed (−6.6%) and for Assignment-focused (+6.5%). The most common tactic to start a course with is Mixed for the first and last quarters and Assignment-focused for the second and third (Appendix D).

2. **Resource-oriented strategy**:

Halves. The Resource-oriented cluster is the second in size (26.6%), with the highest increase in frequency between halves (+6.8%). This is the only strategy cluster to register an actual rise in the number of sessions between halves (+3.4). Compared across the course halves, these students show a remarkable increase in the percentage of Material-focused sessions (4.8%), which was already high in the first half of the course (55.8%). The Mixed tactic is the most common initial session in the first half (59.4%), with Material-focused (39.7%) in the second half. Regarding transition differences, the highest increase is Material-focused→Material-focused (+7%), with the largest decrease being Assignment-focused→Material-focused and Mixed→Mixed (−6%).

Quarters. This cluster is the third in size (25.8%), with small changes initially but a large increase for 3 → 4 (+15.4%). The Material-focused tactics see the biggest increase for 3 → 4 (+7.7%) with Mixed having the highest decrease for 1 → 2 (−7.7%). The Quiz-focused sees very small fluctuations, while the Assignment-focused tactics barely change in the beginning and middle while showing a large decrease for 3 → 4 (+6.7%). The most common initial tactic is Mixed for the first quarter and Material-focused for the rest.

3. **Grade-oriented strategy**:

Halves. The third strategy cluster is the largest (29.1%) with a frequency change of +3.2%; the number of sessions drops by 3.5 between halves. The Assignment-focused tactics represent more than half of the total sessions in both halves (51.5% and 52%, respectively). The Mixed tactic (61.3%) is the most common initial session in the first half, with the Assignment-focused (70.3%) in the second half. The largest transition differences are Material-focused→Material-focused (+8%) and Material-focused→Assignment-focused (−7%).

Quarters. The Grade-oriented strategy is the second in size (27%), with the highest increase in frequency overall for 1 → 2 (+24.3%), barely any change for 2 → 3 (+0.1%), and a large decrease for 3 → 4 (+19.7%). The same patterns can be observed for the average number of sessions per student: an increase for 1 → 2 (+4), small changes for 2 → 3 (+0.5), and a large decrease for 3 → 4 (−10.4). The biggest changes in tactics are the Assignment-focused tactic increase (+6.7%) and Mixed decrease (−6.1) for 1 → 2. This cluster also has the smallest changes regarding the Quiz-focused sessions. Mirroring the previous cluster, the most common initial tactic is Mixed for the first quarter and Assignment-focused for the other three.

4. Flexible strategy:

Halves. The Flexible cluster is the smallest in size (21.7%) with the biggest frequency change between halves (−12%); this cluster also has the highest decrease in the average number of sessions (−5). The Mixed tactic is the most used in both halves, followed by Assignment-focused and Material-focused. This is the only cluster for which the Mixed tactic is the most common initial session in both halves (73.3% and 49.5%, respectively). Finally, this cluster registers the largest transition differences with Quiz-focused→Quiz-focused (+25%) and Quiz-focused→Mixed (−20%).

Quarters. The Flexible cluster is the most often used strategy when including only those students who have been active in all quarters (27.8%) while also seeing the largest change between quarters overall, a decrease of −30.8% for 1 → 2, barely changing in frequency otherwise. The average number of sessions per student slightly increases for the first two changes while significantly decreasing for 3 → 4 (−9). This cluster has the smallest variations in terms of tactic changes as both Material-focused and Quiz-focused tactics barely fluctuate. The largest decrease can be found for the Assignment-focused for 3 → 4 (−5.7%), and the largest increase for Mixed for 3 → 4 (+5.8%). Finally, the most common way to start a course is with the Mixed for all quarters.

The strategies exhibit distinct transitions dependent on the selected periods. The increase in granularity helps interpret the findings. For example, the Balanced strategy demonstrates high consistency in frequency when looking at halves. However, the data across quarters reveals fluctuations, notably a decline in usage midway through the courses followed by a subsequent increase. We can also see that the Assignment-focused tactics are used less in the latter half, with a significant portion of the decrease occurring in the final quarter. Similarly, the number of students using the Resource-oriented strategies escalates during the second half. However, the quarter data indicates this escalation occurs mainly between the third and final quarters. Both Grade-oriented and Flexible strategies see a decrease in the average number of sessions during the latter half; quarterly data highlights that this mostly happens in the last quarter, which is very close to the examination period. Such differences underscore the importance and relevance of using multiple periods when looking at transitions, with different granularity levels contributing smaller pieces to a larger puzzle.

4.3. RQ3: what is the relation between the strategies and strategy transitions on the one hand and final course grades on the other hand, and what are the consequences of maintaining the same strategy versus altering it?

Whole course. A Kruskal-Wallis test indicates that the four strategies significantly differ in their average course grades $H(3) = 27$, $p < .001$. We ran pairwise Dunn tests as post-hoc analyses (using Bonferroni corrections). The results show that the Grade-oriented cluster (Mdn = 6.40, SD = 1.48, n = 1692) scored significantly lower than Balanced (Mdn = 6.63, SD = 1.56, n = 1313; $Z = -3.60$, $p = .002$), Flexible (Mdn = 6.60, SD = 1.79, n = 1404; $Z = -3.73$, $p = .001$) and Resource-oriented students (Mdn = 6.70, SD = 1.78, n = 1461; $Z = -4.70$, $p < .001$). No other significant differences were found.

Course Halves. Kruskal-Wallis tests conducted for each course half suggest a significant difference between strategies when it comes to the final grade: $H(3) = 49$, $p < .001$, and $H(3) = 22$, $p < .001$, respectively. Subsequent post-hoc analyses show that for the first half, the only significant differences were between the Grade-oriented cluster with each of Balanced ($Z = -4.93$, $p < .001$), Flexible ($Z = -4.75$, $p < .001$), and Resource-oriented ($Z = -6.55$, $p < .001$). Conversely, during the second half, significant cluster differences are found between Balanced and Resource-oriented ($Z = -3.65$, $p = .002$) and between Grade-oriented and Resource-oriented ($Z = -3.89$, $p < .001$), the Resource-oriented students scoring higher grades in both cases.

There are three transitions with a frequency above 10%: Grade-oriented→Grade-oriented (15.6%), Resource-oriented→Resource-oriented (15.5%), and Balanced→Balanced (13.5%). The Flexible→Flexible transition also has a high frequency (8.8%), leading to 53.4% of students being Consistent (having the same strategy for both course halves). When comparing the Consistent with Inconsistent students, the Consistent students (Mdn = 6.62, SD = 1.53, n = 2985) scored significantly higher than the Inconsistent students (Mdn = 6.52, SD = 1.69, n = 2606) at course halves level, with $t(5074) = 3.6$, $p < .001$. A further look at the strategies Consistent students used, we also see that Consistent Grade-oriented students score significantly lower than Balanced ($Z = -3.96$, $p < .001$), Flexible ($Z = -5.27$, $p < .001$), and Resource-oriented ($Z = -6.37$, $p < .001$), with no other significant differences.

When comparing the Inconsistent students with each Consistent group, we found the former to score significantly lower than both Consistent Flexible ($Z = -3.69$, $p = .002$) and Resource-oriented ($Z = -4.82$, $p < .001$) students but scored significantly higher than the Consistent Grade-oriented students ($Z = 2.96$, $p = .031$); no significant difference was found with the Consistent Balanced cluster.

Course quarters. For reasons of brevity, we present the most relevant of a larger set of results. Kruskal-Wallis tests suggest significant differences between strategy clusters regarding the final grades for all quarters. Overall, the most consistent finding was that the Grade-oriented cluster scored significantly lower than the other three clusters in the first three quarters (these being the only significant relations for the second and third quarters). The other substantial findings were that Flexible scored significantly lower than Resource-oriented ($Z = -3.38$, $p = .004$), while Balanced scored significantly lower than both Grade-oriented ($Z = -3.65$, $p = .002$) and Resource-oriented ($Z = -5.09$, $p < .001$) in the fourth quarter.

The most common strategy transition was the Consistent Resource-oriented (students that used the Resource-oriented cluster for all four quarters) with 5.9% of all students; 13.5% of students overall are in the Consistent group. Using the same protocol, we compared Consistent, Inconsistent, and Undecided (students that used all four strategies throughout a course) student groups and found that they significantly differed in final grades ($H(2) = 13$, $p = .001$). Similar to the course halves analysis, the Consistent group (Mdn = 6.80, SD = 1.51, n = 691) scored significantly higher than both the Inconsistent (Mdn = 6.60, SD = 1.45, n = 4332; $Z = 3.08$, $p = .006$) and Undecided (Mdn = 6.40, SD = 1.49, n = 87; $Z = 2.73$, $p = .019$) groups; the Inconsistent and Undecided cluster did not significantly differ in their grades.

When comparing the Consistent students between themselves based on the only strategy they used, the Grade-oriented students (Mdn = 6.00, SD = 1.41, n = 110) scored significantly lower than both Resource-oriented (Mdn = 7.07, SD = 1.52, n = 299; $Z = -6.65$, $p < .001$) and Consistent Flexible students (Mdn = 7.1, SD = 1.58, n = 123; $Z = -5.97$, $p < .001$); similarly, the Consistent Balanced

students (Mdn = 6.62, SD = 1.34, n = 159) scored significantly lower than both the Resource-oriented ($Z = -3.92$, $p < .001$) and the Flexible Consistent students ($Z = -3.56$, $p = .003$), with no significant difference when compared to the Consistent Grade-oriented cluster. Finally, when comparing the Inconsistent students with each Consistent group, we found the former to score significantly lower than both Consistent Flexible ($Z = -4.06$, $p < .001$) and Resource-oriented ($Z = -5.50$, $p < .001$) students but scored significantly higher than the Consistent Grade-oriented students ($Z = 4.27$, $p < .001$); no significant differences were found when compared with the Consistent Balanced cluster.

In summary, our study reveals that students who consistently utilized specific learning strategies throughout various course time periods achieved significantly higher grades than those who changed their strategies at any point. Moreover, adhering consistently to deep learning methods resulted in superior grades compared to a consistent adoption of surface learning techniques (the Grade-oriented strategy). Furthermore, students displaying inconsistency in their approach attained notably higher scores than those consistently employing surface learning strategies. These findings prove that it is best to start and continually use deep learning approaches, however, combining approaches is also better than using only surface learning strategies.

5. Discussion

We investigated changes in online behavior patterns in higher education students based on trace data from two years and a total of 41 courses and 5870 students, a scale that, to our knowledge, has not been analyzed before. We identified four different tactics that students used within the courses, which led to identifying four strategies in the whole course data, for each course half, and for each course quarter. The final grades showed differences based on the strategies used but also based on the consistency of their employment across the course periods.

5.1. RQ1: which learning tactics and strategies of students can be identified through the analysis of the LMS's trace data?

The observed online engagement patterns reveal notable trends across learning tactics. Course browsing and engagement with Learning materials emerge as predominant actions, underlining their fundamental role in learners' interactions within the LMS. The predominant tactic is Assignment-focused, which describes sessions with the most clicks on Mandatory quizzes and Assignment pages. As expected, this tactic also has the highest number of clicks on Submission, which suggests that these sessions are used for fulfilling course requirements and meeting assessment criteria, underscoring students' deliberate and targeted approach toward task-oriented behaviors. The second most common tactic is Material-focused, intuitively obvious given the nature of the LMS, which is often used as a repository for course material. Engagement with Learning materials reflects an initiative toward resource utilization and information processing, a fundamental aspect of SRL, where learners set goals and plan activities to achieve academic objectives (the first and second phases of the COPES model (Winne & Hadwin, 1998). The third most common tactic is the Mixed tactic, which manifests a versatile and diversified engagement within the LMS. Unlike the preceding tactics, this cluster encompasses a more balanced distribution of actions within sessions, emphasizing information acquisition and metacognitive engagement, an aspect crucial in SRL for monitoring and regulating learning processes (Panadero, 2017). Almost no initial clicks on the Submission action suggest a preliminary phase akin to SRL's forethought phase from Zimmerman's model (Zimmerman, 2008) or the task definition phase from the COPES model (Winne & Hadwin, 1998), wherein learners gather general information about the course before delving into substantive learning materials. A general Mixed tactic that does not focus on a specific LMS feature and is used by students to direct their general learning process has also been identified in many other studies, suggesting that their existence is necessary and present across environments and contexts (e.g., Lim, Gasevic, Matcha, Ahmad Uzir, & Dawson, 2021, pp. 364–374; Maldonado-Mahauad, Pérez-Sanagustín, Kizilcec, Morales, & Munoz-Gama, 2018). The least frequently observed tactic among learning sessions, the Quiz-focused tactic, has the longest average session length, indicating learners' sustained and concentrated effort. The dominance of quiz-related clicks, especially Mandatory quizzes, signifies an engagement with assessment tasks, similar to the Assignment-focused tactic, aligning with performance-oriented activities and self-assessment.

Our analyses identified some learning strategies that fit in the deep versus surface learning (Entwistle, 1991) or in the goal orientation paradigms (Jovanović et al., 2017; Kovanović et al., 2019; Payne, Youngcourt, & Beaubien, 2007; Steinmayr, Bipp, & Spinath, 2011), although not perfectly. We labeled these strategies as Balanced, Grade-oriented, Resource-Oriented, or Flexible. The Balanced students showed a nearly equal distribution of tactics between Material-focused, Assignment-focused, and Mixed, with the highest usage of the Quiz-focused tactic. This cluster is similar in composition to the Grade-oriented strategy, however, students replacing some of the assignment usage with quizzes. This pattern could suggest indecisiveness in learning behavior, as students oscillate between reviewing course objectives and studying materials to achieve those objectives. Conversely, deep learning strategies and mastery goal orientation prioritize comprehension and critical thinking, emphasizing genuine learning over grades (Biggs, Kember, & Leung, 2001; Dolmans, Loyens, Marcq, & Gijbels, 2016; Hattie & Donoghue, 2016; Payne et al., 2007). Resource-oriented students demonstrate a focused approach primarily centered on the Material-focused tactic, behavior akin to a deep approach to learning. Such students spend most of their time accessing and hopefully studying the available material, the actual assessment criteria and grading becoming a secondary thought.

While the previous strategies clearly fit into traditional learning strategy paradigms, the Flexible cluster is not as clear. Hattie and Donoghue (2016) argue that the adaptation and transition between different contexts constitute a strategy itself, known as the transfer strategy. Similarly, the strategic approach to learning is associated with students changing between surface and deep learning strategies in accordance with the task at hand (Richardson, Abraham, & Bond, 2012). It is possible that Flexible students defy the deep vs. surface learning dichotomy by employing a variety of tactics that facilitate seamless transfer between course periods, aligning with the

mentioned strategies. This approach may resemble the Mixed tactic, which is typically utilized for gaining an overview of course objectives and materials before students commit to either a surface or deep learning strategy (Winne & Hadwin, 1998; Zimmerman, 2008).

5.2. RQ2: which strategy transitions can be distinguished between the halves and quarters of the course, and what characteristics define these transitions?

Looking at the data based on course halves, we see a slight increase between the first and second half in the number of Grade-oriented strategies. This is somewhat expected, as most courses have a final exam or assignment, which students would access closer to their deadline in a performance-oriented behavior. However, examining the data in quarters shows that the Grade-oriented strategy peaks in the middle of the course, followed by a notable decline during the exam period. One possible explanation is the presence of interim assignments commonly found in the studied courses. Jovanović et al. (2017) describe a similar learning profile, with students being selective and efficient.

Mirroring this trend, Resource-oriented strategies exhibit a decline during the middle of the course, followed by a notable increase in usage during the final quarter, suggesting a transition towards more effective strategies in the concluding weeks of courses. The average number of sessions per period serves as a proxy for engagement, revealing an increase in sessions for all strategies during the middle portion of the course, followed by a decrease in the last part. While this may initially appear as a decline in engagement, indicative of students spending less time on learning, it could also represent an increase in preparation activities outside the LMS. It could also reflect a significant improvement in efficiency, as suggested by Jovanović et al. (2017). They argue that, early in the course, students spend more time accessing various LMS features to comprehend course criteria and materials thoroughly. As the course progresses and the final deadline looms nearer, they streamline their behavior, focusing primarily on relevant information within the LMS. Although all strategies converge towards minimal online engagement by the course's end, students implement this approach in diverse ways using different tactics. Research on the successful transfer of deep strategies across courses presents conflicting findings compared to studies on surface strategies, warranting further investigation in future studies (Asikainen & Gijbels, 2017; Saqr et al., 2023a, b, c).

Another intriguing observation is that more than half of all students initially adopt the Flexible strategy in the first quarter, characterized by the shortest sessions overall, followed by a significant decline in subsequent quarters. This behavior aligns with findings by Fan et al. (2022), who suggest that this initial surge in Flexible strategy usage could represent an initial phase of material exploration before students settle into a more sustainable strategy. We posit that this trend may indicate a form of SRL, where students employ various learning techniques initially and gradually transition to a more suitable and effective learning strategy over time (Greene & Azevedo, 2007). Similarly, the Balanced strategy displays the lowest utilization, with consistent frequency observed across halves and quarters, accompanied by the longest number of sessions per course. This consistency suggests that students adopting the Balanced strategy maintain a steady approach, rarely altering their behavior between course periods. These students meticulously engage with all LMS features throughout the course, irrespective of approaching deadlines, demonstrating a consistent learning approach even during interim or final exams.**5.3. RQ3: What is the relation between the strategies and strategy transitions on the one hand and final course grades on the other hand, and what are the consequences of maintaining the same strategy versus altering it?**

As noted by Saqr and López-Pernas (2021), there is an association between students' online consistency and achievement, a relation also discovered in this paper. In assessing academic performance, the adoption of a surface learning paradigm, here reflected in the Grade-oriented strategy, was associated with notably inferior grades relative to the other three strategies. This has been identified as well in previous studies, such as Fincham et al. (2019) and Fan, Matcha, Uzir, Wang, and Gašević (2021). During the second half of the course, adopting a deep learning paradigm, here reflected in the Resource-oriented strategy, yielded significant performance improvement compared to surface learning approaches (both Grade-oriented and Balanced strategies). This demonstrates students' capacity for successful regulation (Fincham et al., 2019) and emphasizes that transitioning into a deep learning paradigm can help. However, the grades of those transitioning into deep learning were still lower than those of students using deep learning strategies throughout. Fan et al. (2022) found the same performance improvement, as students who applied a surface learning strategy initially received higher grades when applying a deep learning strategy during their retake. In this sense, we do not find evidence for arguments suggesting that surface learning is a necessary component of a successful learning strategy.

The adoption and continued use of specific strategies across temporal intervals generally resulted in better performance compared to frequent strategy changes, aligning with prior research indicating an advantage for students who maintain certain learning approaches (Saqr & López-Pernas, 2021b; Saqr et al., 2023a, b, c). Then again, being consistent is not enough, as students consistently choosing surface approaches scored the lowest performance, even when compared to Inconsistent students. Similar results were reported by Saqr et al. (2023a, b, c), who showed that students may face challenges adapting deep strategies to new course requirements. This disparity has been suggested to be due to the potential increase in cognitive load associated with such adaptations (Winne &

Jamieson-Noel, 2003). More research is required to determine why some students struggle to transition out of surface-based strategies. Our findings are in line with the learning model proposed by Hattie and Donoghue (2016), one of the foundations for our study, as students employing deep learning strategies across the entire course faired best. We found it difficult to find evidence for the successful use of transfer strategies (the repeated use of deep learning in other contexts). In any case, only a minority of students initiated with a deep learning strategy and maintained it across quarters (5.9%), so if students use transfer strategies, it is relatively rare. This underscores the importance of examining strategy evolution over time, particularly how to assess students applying a certain strategy in a given area to another.

Students who consistently utilized the Flexible strategy exhibited behavior akin to that of the Resource-oriented group, consistently achieving significantly higher scores than the Grade-oriented, Balanced groups and Inconsistent students in general. A possible explanation is that this difference in performance can be attributed to the composition and utilization of learning tactics within each cluster. Although the Flexible and Balanced clusters may appear similar on the surface, there are distinct differences in the tactics employed. Flexible students utilize the most Mixed tactics, characterized by engagement with various LMS features during each session (e.g., accessing discussion forums, lectures, articles, and course criteria), reflecting a diverse and exploratory approach to learning. In contrast, the Balanced strategy encompasses a more varied use of tactics overall, with students distributing their sessions equally between Assignment-focused and Material-focused. This distinction in learning tactics sheds some light on the contrasting outcomes observed among students adopting Flexible and Balanced strategies. While the Flexible strategy allows for a broader exploration of course content and resources, potentially enhancing understanding and retention, the Balanced strategy emphasizes a more even distribution of time and focus across different learning tasks. This nuanced understanding of students' learning behaviors underscores the importance of considering not only the strategies themselves but also the specific tactics employed within each strategy when evaluating their impact on academic performance. Further research exploring the interplay between learning strategies and tactics is warranted to elucidate their implications for student success in online learning environments.

5.3. Limitations and future research

This study examined the evolution of student strategies across multiple periods. To capture a comprehensive pattern, our analysis focused on students who remained engaged throughout all selected course periods, inadvertently introducing a survivorship bias, as highlighted by Saqr et al. (2023a, b, c). This bias stems from the inclusion of students who persisted throughout the entire study duration, potentially leading to results skewed towards a more positive outlook. Notably, we lacked data on students who did not participate in certain course periods, such as those who did not sign in at all during the second quarter, thereby limiting our ability to draw conclusions about their learning strategies and behaviors.

As previously noted, the LMS's inherent nature introduces uncertainties regarding students' precise intentions when performing online actions. For instance, a click on an article may signify either an article being opened, followed by prolonged inactivity or concentrated engagement driven by specific learning objectives. In our analysis, we observed an actual decrease in activity among Resource-oriented students nearing the examination date. While counter-intuitive, this trend could indicate students downloading the learning material and studying offline during the final course weeks. Due to specific course design choices, online engagement metrics may not fully capture the entirety of learning activities. This is further exacerbated by the heterogeneity of classes, learning objectives, materials, student behavior, and even variations in data quality. While it is possible to infer certain "offline" actions from the trace data, this requires a deep understanding of the course context, ideally supplemented with teacher insights. In line with Panadero et al. (2016), we agree that self-reported data can help in such cases by revealing the intention and motivation behind the trace data. While previous studies have acknowledged similar limitations (e.g., Baker et al., 2020; Jovanović et al., 2017), establishing definitive guidelines remains elusive, and future research needs to be done to address this gap comprehensively.

An important strength of our study was the extensive dataset derived from many courses, facilitating the identification of clear patterns in student behavior across diverse contexts. As observed when comparing two studies that used the same sample, Fincham et al. (2019) and Jovanović et al. (2017), including context details in the analysis can lead to different identified learning patterns. In our case, it could be that the Flexible and Balanced clusters were identified due to such conditions that might not resurface in other studies. Such conditions can range from internal (e.g., motivation, self-efficacy, and cognitive competence) to external (available resources, instructional cues, and course goals) (Greene & Azevedo, 2007; Panadero, 2017; Srivastava et al., 2022, pp. 392–403). For example, Heikkinen et al. (2024) found a relationship between internal conditions (i.e., student self-efficacy and goal-setting) and the learning tactics students used. On the other hand, Kleter, Matzat, and Conijn (2024) discuss how even small variations in course setup, content, or instructional cues can impact model generalizability. In our study, the limited detail on course design characteristics remains a constraint, as variations in learning design could influence these learning patterns. This limitation underscores the need for future studies to incorporate comprehensive design data to deepen insights of learning strategies and their contextual dependencies. For example, future studies should include various covariates that arise at both student or course levels, such as the number of assignments, number of lectures, group work presence, or even students' prior knowledge. Including such variables should allow for a more contextualized understanding of learning mechanisms.

Several limitations arise due to this study's data type and methodology. While utilizing unobtrusive data offers certain advantages, such as efficiency and objectivity (Cristea et al., 2023; Panadero et al., 2016), it also imposes limitations on the scope of analysis due to the absence of access to precise motives or emotional aspects (Saqr & López-Pernas, 2021b). As aptly stated by Jovanović et al. (2017), the "why" behind students' behaviors remains unknown. Future studies could benefit from employing multi-modal methods to address this gap and provide a more comprehensive understanding of students' learning experiences. Moreover, while Markov Modelling and clustering techniques enable the discovery of new patterns in online learning, they are inherently black-box techniques, introducing a level of subjectivity, particularly when determining the final clusters (Fincham et al., 2019). Similarly, attributing each student to only one cluster at a time leads to a reductionist approach, which can obscure certain behaviors (Saqr & López-Pernas, 2021b). We must accept this trade-off for now; however, future research should explore methods to mitigate and account for this subjectivity, enhancing the robustness and validity of findings in online learning research.

6. Conclusion

In summary, our study leverages big data to delineate four distinct tactics, subsequently organized into four overarching strategies. This study builds on previous work by specifically comparing these tactics and strategies within courses, across course halves and quarters, and relating this with academic performance.

This comparative analysis offers a more comprehensive understanding of the dynamics of learning strategies. By elucidating the subtle relationship between strategy usage, transitions, and academic performance over time, educators can tailor instructional approaches to optimize student learning outcomes. Future studies can explore how surface and deep learning strategies transfer within and between course periods, aiming to discern potential variations across different contexts. Furthermore, interventions can be developed to acknowledge that surface learning strategies are acceptable in certain contexts, provided that students eventually transition to more effective learning behavior. This nuanced approach to intervention can pave the way for more impactful strategies to enhance students' learning experiences.

CRediT authorship contribution statement

T.S. Cristea: Writing – review & editing, Writing – original draft, Visualization, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **S. Heikkinen:** Writing – review & editing, Software, Methodology, Conceptualization. **C. Snijders:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **M. Saqr:** Writing – review & editing, Software, Methodology, Conceptualization. **U. Matzat:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **R. Conijn:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **A. Kleingeld:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Availability of data and materials

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request. The R scripts used in this article can be found at: Tudor-Cristea/pattern-paper-scripts (github.com).

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Declaration of interests

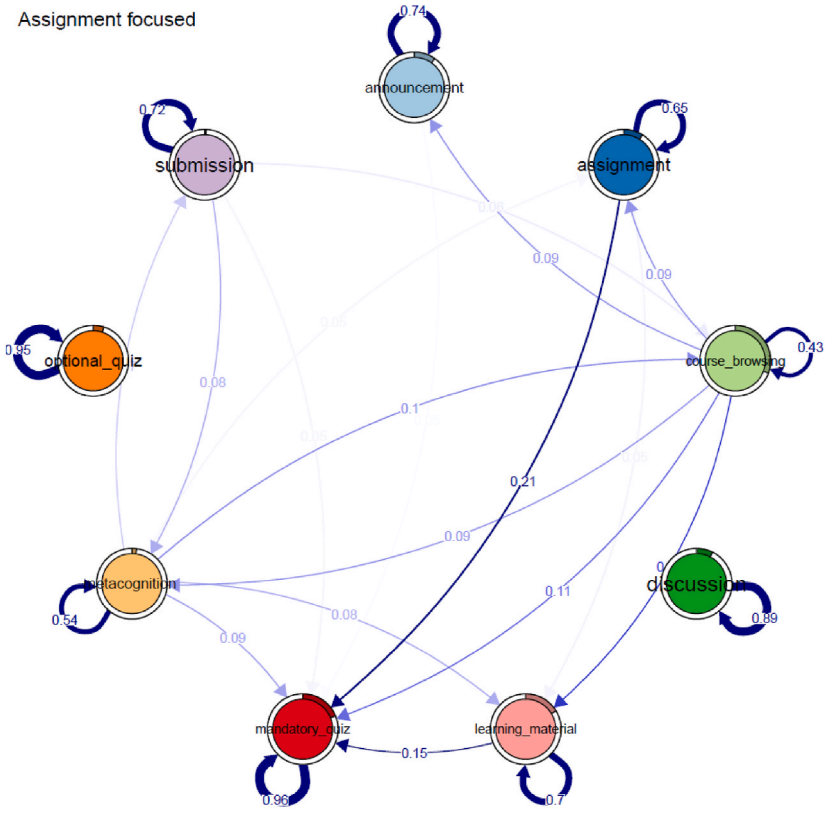
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

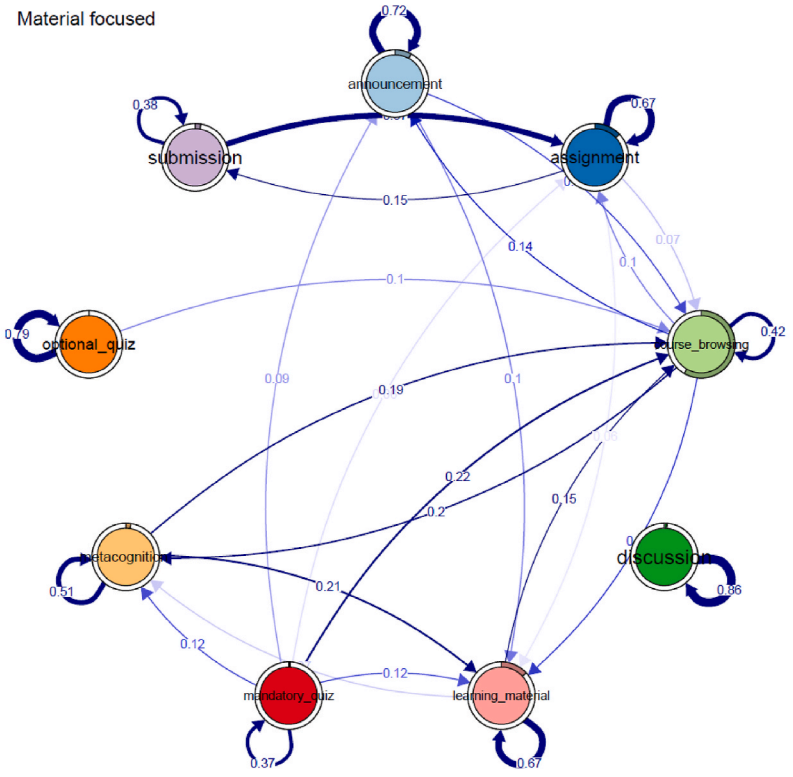
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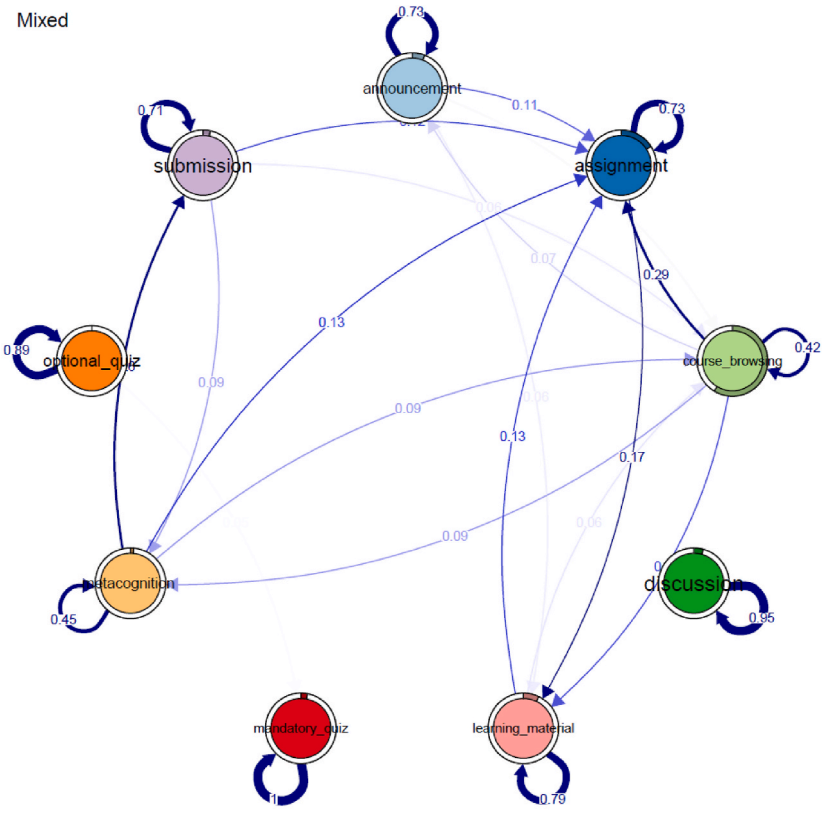
The analysis used the Dutch national e-infrastructure with the support of the SURF Cooperative using grant no. EINF-7346.

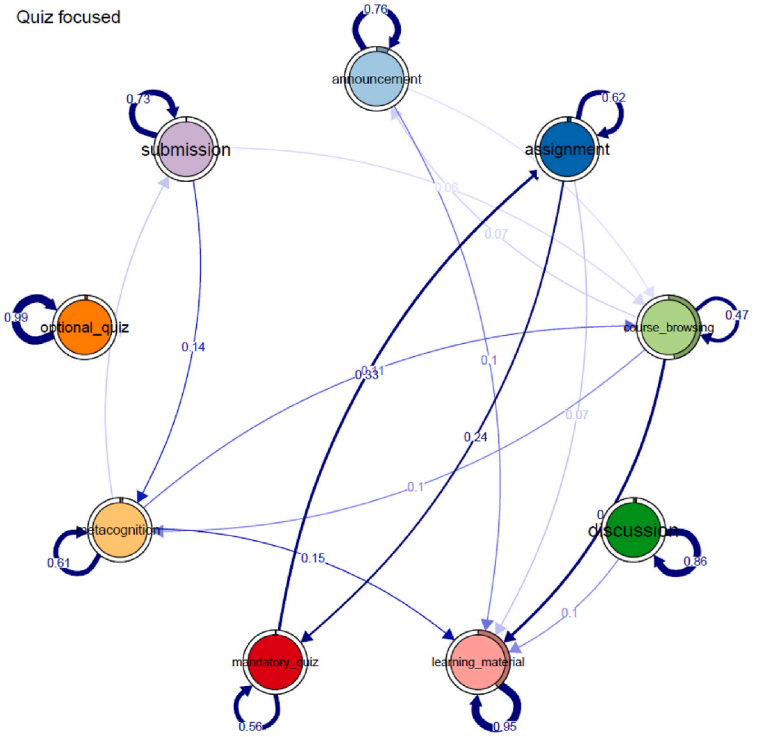
Appendix A

Initial and Transition Probabilities of the Actions per Tactic Cluster.


Assignment focused
announcement
submission
assignment
optional_quiz
course_browsing
metacognition
discussion
mandatory_quiz
learning_material
0.74
0.72
0.65
0.95
0.43
0.54
0.89
0.96
0.7
0.09
0.08
0.1
0.21
0.11
0.15

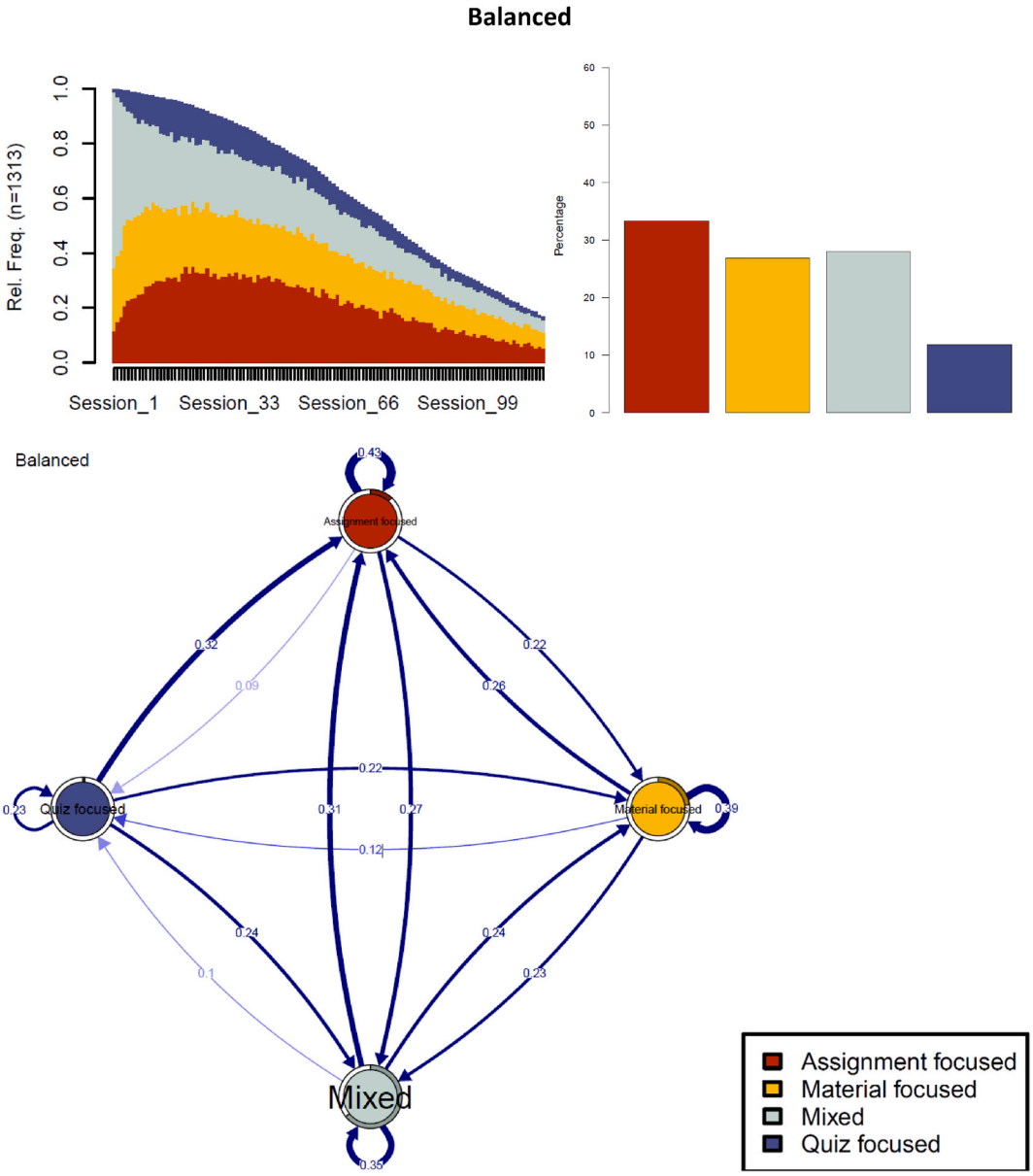

Material focused
announcement
submission
assignment
optional_quiz
course_browsing
metacognition
discussion
mandatory_quiz
learning_material
0.72
0.38
0.67
0.79
0.42
0.51
0.86
0.37
0.67
0.15
0.14
0.1
0.19
0.22
0.2
0.21
0.12

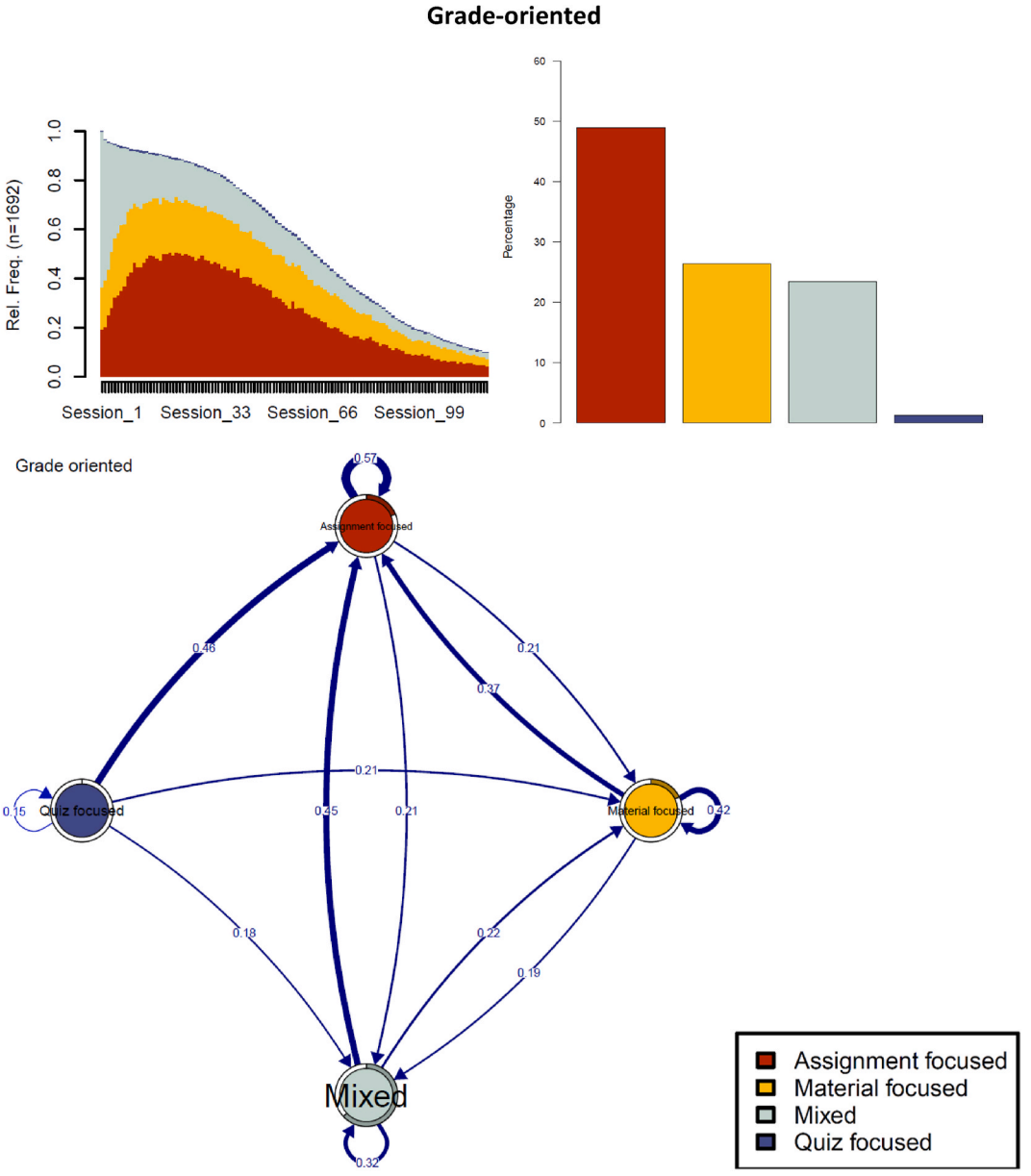


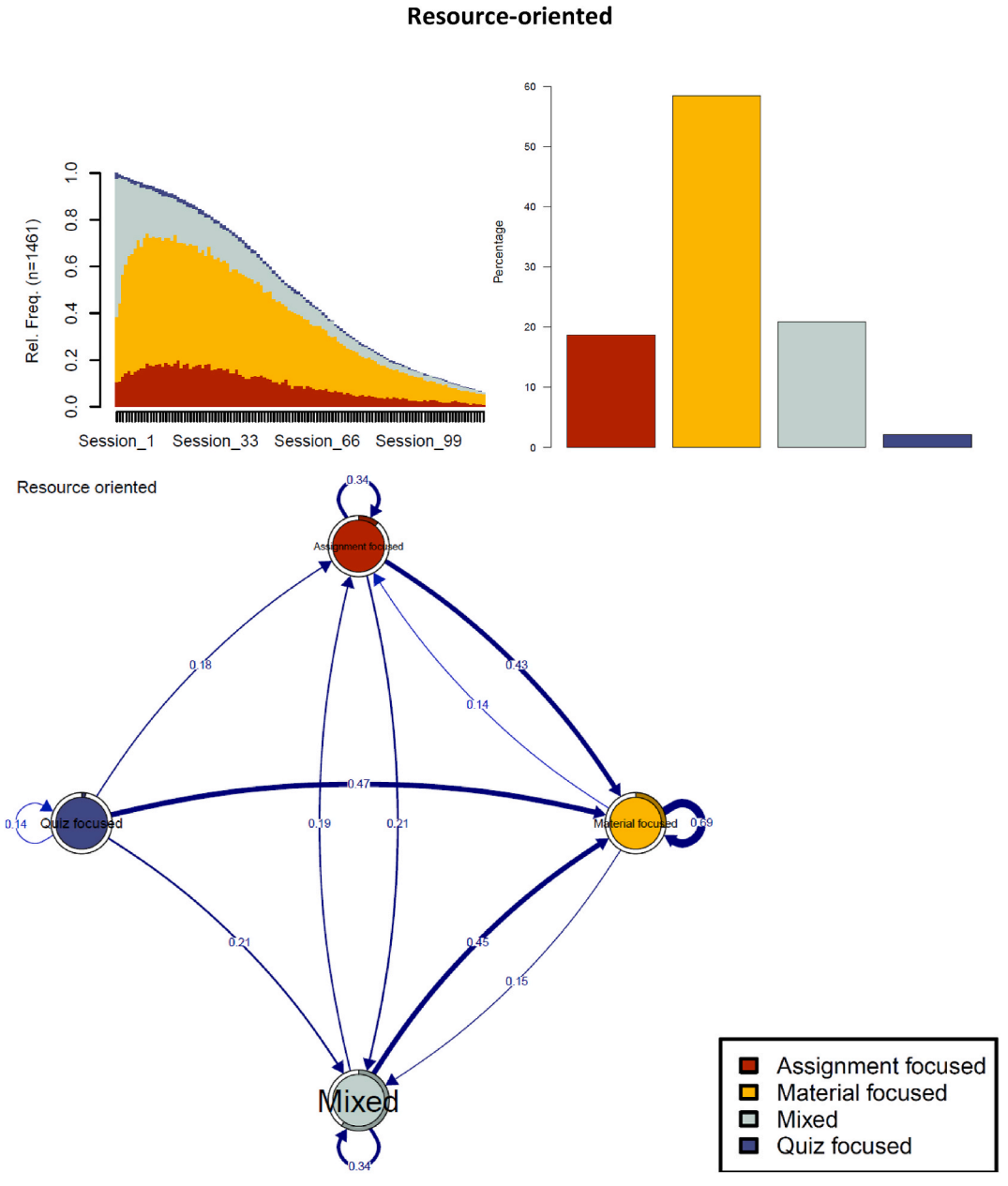


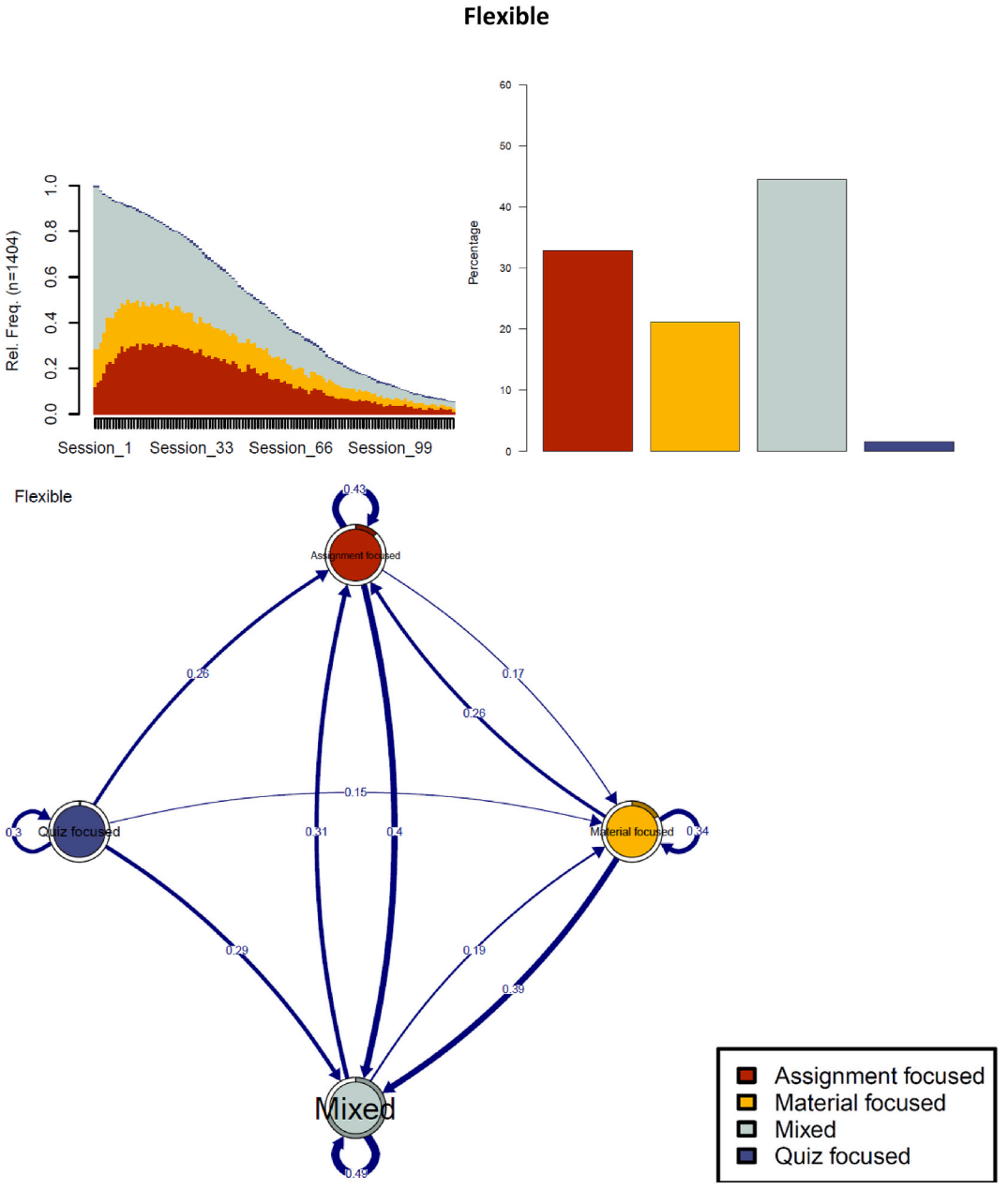
Appendix B

Temporal Distribution, Frequency Marplot, and Transition Probability Plot of the Tactics per Strategy Cluster.


Balanced
Rel. Freq. (n=1313)
0.0
0.2
0.4
0.6
0.8
1.0
Session_1
Session_33
Session_66
Session_99
Percentage
0
10
20
30
40
50
60
Balanced
Assignment focused
Material focused
Quiz focused
Mixed
0.43
0.32
0.09
0.22
0.26
0.22
0.23
0.31
0.27
0.39
0.12
0.24
0.1
0.24
0.23
0.35
Assignment focused
Material focused
Mixed
Quiz focused

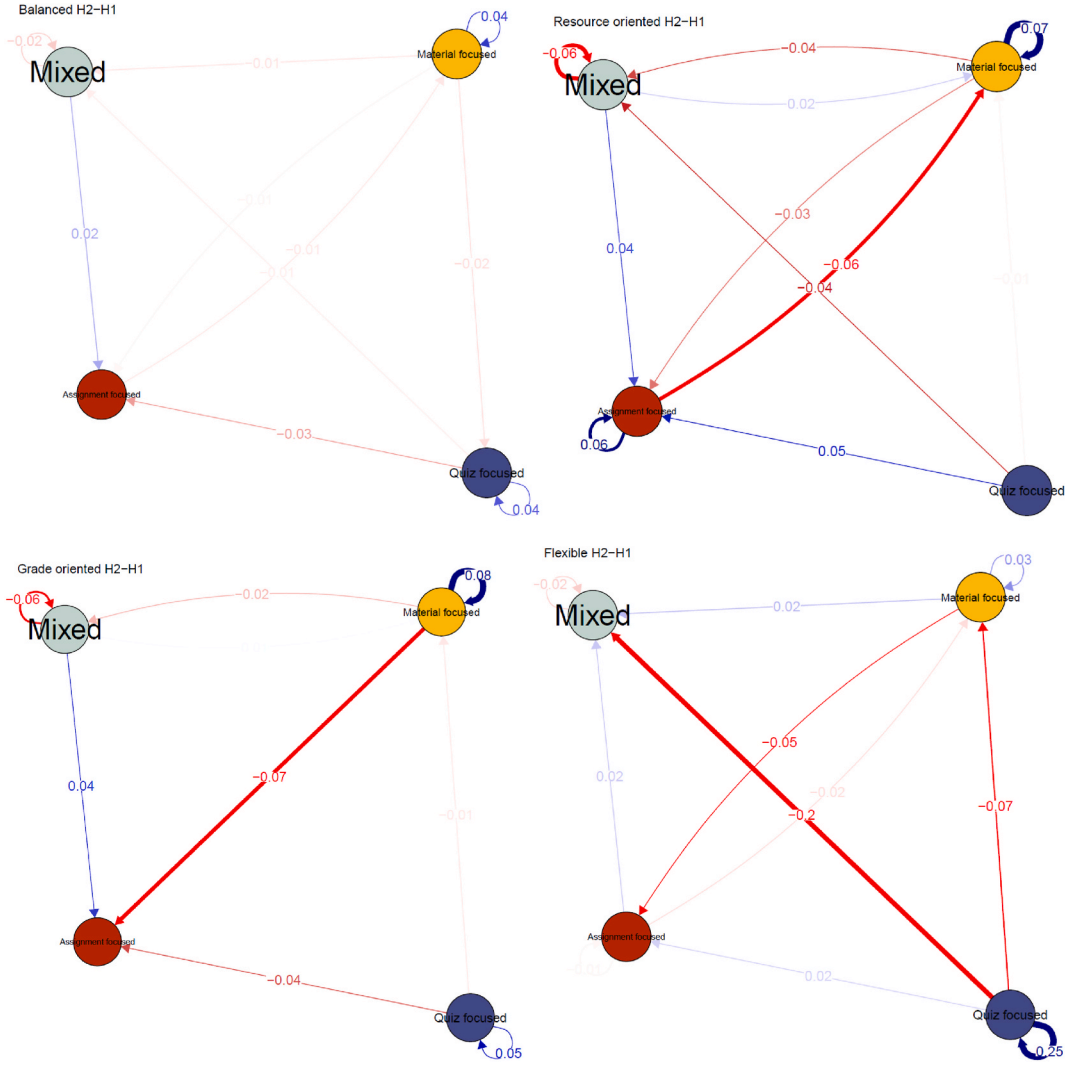

Grade-oriented
Rel. Freq. (n=1692)
0.0
0.2
0.4
0.6
0.8
1.0
Session_1
Session_33
Session_66
Session_99
Percentage
0
10
20
30
40
50
60
Grade oriented
Assignment focused
Quiz focused
Material focused
Mixed
0.57
0.46
0.21
0.37
0.21
0.15
0.45
0.21
0.42
0.18
0.22
0.19
0.32
Assignment focused
Material focused
Mixed
Quiz focused


Resource-oriented
Rel. Freq. (n=1461)
0.0
0.2
0.4
0.6
0.8
1.0
Session_1
Session_33
Session_66
Session_99
Percentage
0
10
20
30
40
50
60
Resource oriented
Assignment focused
Quiz focused
Material focused
Mixed
0.34
0.18
0.43
0.14
0.47
0.14
0.19
0.21
0.69
0.21
0.45
0.15
0.34
Assignment focused
Material focused
Mixed
Quiz focused


Flexible
Rel. Freq. (n=1404)
0.0
0.2
0.4
0.6
0.8
1.0
Session_1
Session_33
Session_66
Session_99
Percentage
0
10
20
30
40
50
60
Flexible
Assignment focused
Quiz focused
Material focused
Mixed
0.43
0.26
0.17
0.26
0.15
0.3
0.31
0.4
0.34
0.29
0.19
0.39
0.49
Assignment focused
Material focused
Mixed
Quiz focused

Appendix C

Transition Probability Difference Plots per Strategy Between Course Halves. Blue represents an increase, while red represents a decrease from the first half to the second.



Appendix D

Table with Descriptive Data of the Four Strategies for the Four Quarters.

Data	Balanced (n = 3969)				Resource-oriented (n = 5277)				Grade-oriented (n = 5517)				Flexible (n = 5677)			
	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
Frequency	648 (12.7%)	1138 (22.3%)	1006 (19.7%)	1177 (23%)	1202 (23.5%)	1050 (20.5%)	1118 (21.9%)	1907 (37.3%)	702 (13.7%)	1935 (37.9%)	1943 (38%)	937 (18.3%)	2558 (50.1%)	987 (19.3%)	1043 (20.4%)	1089 (21.3%)
Average number of sessions per student per course (SD)	19.4 (8.9)	22.8 (9.4)	21.3 (9.1)	15.9 (8.6)	15.4 (9.3)	19.5 (10.1)	21.3 (10.1)	17.1 (8.6)	15.8 (8.9)	19.8 (9.1)	20.3 (9.8)	9.9 (7.3)	14.8 (9.1)	18.7 (9.4)	18.8 (9.5)	9.8 (6.9)
Assignment-focused	30%	36.5%	36%	29.3%	17.8%	20%	21.2%	14.9%	52.3%	59%	58.9%	57.9%	25.8%	29.2%	29.5%	23.8%
Material-focused	26.3%	23.3%	24.1%	25.8%	55.8%	60.4%	59.3%	67%	20.8%	20.1%	19.1%	20.6%	21.7%	22%	22.8%	22.4%
Mixed	31.2%	24.6%	24.6%	26.6%	24.6%	16.9%	17.4%	15.5%	26.5%	20.4%	21.3%	21%	51.5%	48.2%	46.7%	52.5%
Quiz-focused	12.5%	15.6%	15.3%	18.3%	1.8%	2.7%	2.1%	2.6%	0.4%	0.5%	0.7%	0.5%	1%	0.6%	1%	1.3%
Most common first tactic in a course	Mixed (59.6%)	Assignment-focused (32%)	Assignment-focused (34.7%)	Mixed (36.4%)	Mixed (55.2%)	Material-focused (48.1%)	Material-focused (44.6%)	Material-focused (48.9%)	Mixed (45.4%)	Assignment-focused (67%)	Assignment-focused (73.1%)	Assignment-focused (73%)	Mixed (76.2%)	Mixed (57.1%)	Mixed (54%)	Mixed (59.4%)

Appendix E

Descriptive Data of the Four Strategies for the Course Quarters and the Transition Between Them.

	Balanced (n= 3969, 19.4%)			Resource-oriented (n= 5277, 25.8%)			Grade-oriented (n= 5517, 27%)			Flexible (n= 5677, 27.8%)		
Data	1→2	2→3	3→4	1→2	2→3	3→4	1→2	2→3	3→4	1→2	2→3	3→4
Frequency	+490 (9.6%)	-132 (2.6%)	+171 (3.3%)	-152 (3,0%)	+68 (1.4%)	+789 (15.4 %)	+1233 (24.2 %)	+8 (0.1%)	-1006 (19.7 %)	-1571 (30.8 %)	+56 (1.1%)	+46 (0.9%)
Average number of sessions per student per course (SD)	+3.4 +(0.5)	-1.5 -(0.3)	-5.4 -(0.5)	+4.1 +(0.8)	+1.8 (0.0)	-4.2 -(1.5)	+4.0 (+0.2)	+0.5 +(0.7)	-10.4 -(2.5)	+3.9 +(0.3)	+0.1 +(0.1)	-9.0 -(2.6)
Assignment-focused	+6.5%	-0.5%	-6.7%	+2.2%	+1.2%	-6.3%	+6.7%	-0.1%	-1.0%	+3.4%	+0.3%	-5.7%
Material-focused	-3.0%	+0.8%	-1.7%	+4.6%	-1.1%	+7.7%	-0.7%	-2.0%	+1.5%	+0.3%	+0.8%	-0.4%
Mixed	-6.6%	0.0%	+2.0%	-7.7%	+0.5%	-1.9%	-6.1%	+0.9%	-0.3%	-3.3%	-1.5%	+5.8%
Quiz-focused	+3.1%	-0.3%	+3.0%	+0.9%	-0.6%	+0.5%	+0.1%	+0.2%	-0.2%	-0.4%	+0.4%	+0.3%

Note: Green represents an increase, while red represents a decrease between the course quarters.

Data availability

I have shared a link to the code used in the Method section.

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